

JEE

ADVANCED 2024



Physics

FINAL PRACTICE
PAPER-1



HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PHYSICS

SECTION-I : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

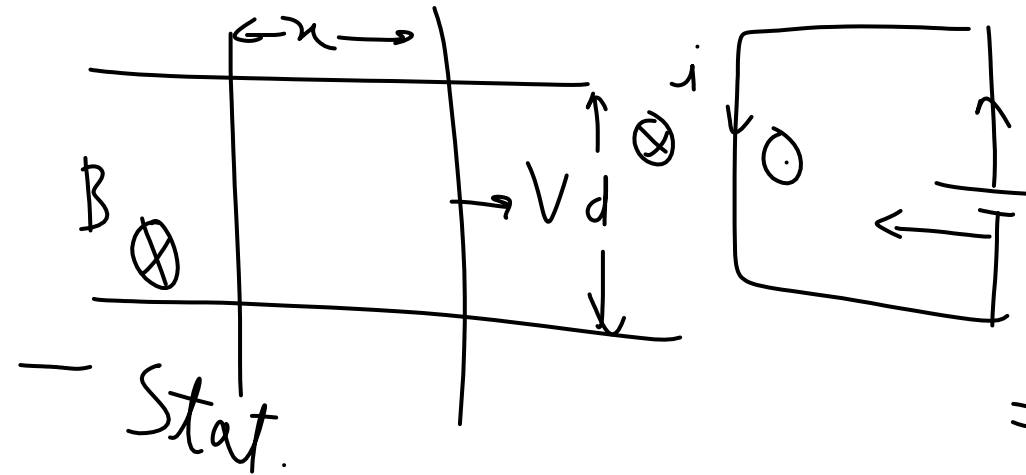
1. Two long wires are placed on a pair of fixed parallel rails perpendicular to the wires. The spacing d between the rails is large compared with x , the distance between the wires. Both wires and rails are made of a material of resistance ρ per unit length. A magnetic flux density B is applied perpendicular to the rectangle so formed, and one wire is moved along the rails with a uniform speed v while the other is held stationary. It is seen that the force required to hold the stationary wire vanishes at a particular value of x . Find x .

(A) $\frac{\mu_0 v}{4\pi\rho}$

(B) $\frac{\mu_0 v}{2\pi\rho}$

(C) $\frac{2\mu_0 v}{\pi\rho}$

(D) $\frac{\mu_0 v}{3\pi\rho}$



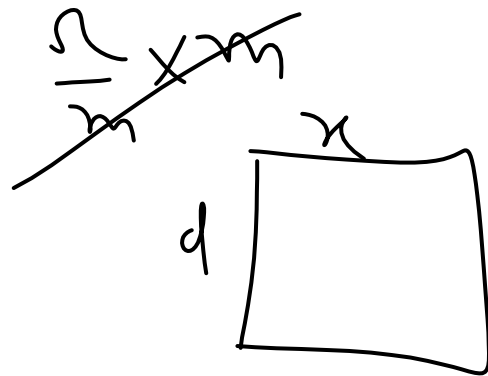
$$B = \frac{\mu_0 i}{2\pi x}$$

$$= \frac{\mu_0}{\pi x} \left(\frac{B v l}{R} \right)$$

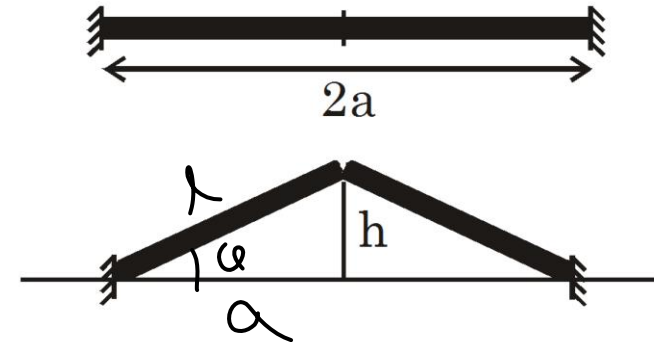
$$B = \frac{\mu_0}{2\pi r} \left(\frac{B V l}{R} \right) = \frac{\mu_0}{2\pi r} \frac{B V d}{\rho(d+r)} \Rightarrow 1 = \frac{\mu_0 V d}{4\pi r \rho d} \quad d \gg r$$

$$\therefore \frac{R}{l} = \frac{\rho}{A}$$

$$\Rightarrow r = \frac{V \mu_0}{4\pi \rho}$$



2. A bar with a crack at its centre buckles as a result of temperature rise of ΔT . If the fixed distance is $2a$ and the coefficient of linear expansion of bar is α , find the rise h of centre.



(A) $a\sqrt{\alpha^2\Delta T^2 + \alpha\Delta T}$

(B) $a\sqrt{\alpha^2\Delta T^2 - \alpha\Delta T}$

☒ (C) $a\sqrt{\alpha^2\Delta T^2 + 2\alpha\Delta T}$

(D) $a\sqrt{\alpha^2\Delta T^2 - 2\alpha\Delta T}$

Sol $\cos \theta = \frac{a}{l} = \frac{a}{a(1 + \alpha\Delta T)} = \frac{1}{1 + \alpha\Delta T} \quad (i)$

$\tan \alpha = \frac{h}{a} \Rightarrow h = a \tan \theta$

$$h = a \tan u = a \sqrt{(1 + \alpha \cot)^2 - 1} = a \sqrt{1 + \alpha^2 \cot^2 + 2\alpha \cot - 1}$$

$$\cos u = \frac{1}{1 + \alpha \cot} \quad B \quad = a \sqrt{\alpha^2 \cot^2 + 2\alpha \cot}$$

$$\tan u = \frac{P}{B} = \frac{\sqrt{(1 + \alpha \cot)^2 - 1}}{1}$$

3. The power output from the motor on a 2 kg body depends on the velocity of body as $P = v(5-v)$. The work done on the body for first 2 second is

(A) $25(e-1)$

(B) $5\left(\frac{e-1}{e}\right)$

☒ (C) $25\left(\frac{e-1}{e}\right)^2$

(D) $25\left(\frac{e^2-1}{e^2}\right)^2$

Sol $m = 2 \text{ kg}$, $P = v(5-v)$

$$Fv = v(5-v)$$

$$F = 5-v$$

$$ma = 5-v$$

$$a = \frac{5-v}{2}$$

$$\frac{dv}{dt} = \frac{5-v}{2}$$

$$\int_0^v \frac{dv}{5-v} = \int_0^t \frac{dt}{2}$$

$$-\ln(5-v) = \frac{t}{2}$$

$$\left(\ln \left(\frac{S-v}{S-u} \right) \right)^V = -\frac{\pi}{2}$$

$$\ln \left(\frac{S-v}{S-u} \right) = -\frac{\pi}{2}$$

$$\frac{S-v}{S-u} = e^{-\pi/2}$$

$$t=0, u=0$$

$$\frac{S-v}{S} = e^{-\pi/2}$$

$$S-v = S e^{-\pi/2}$$

$$v = S - S e^{-\pi/2}$$

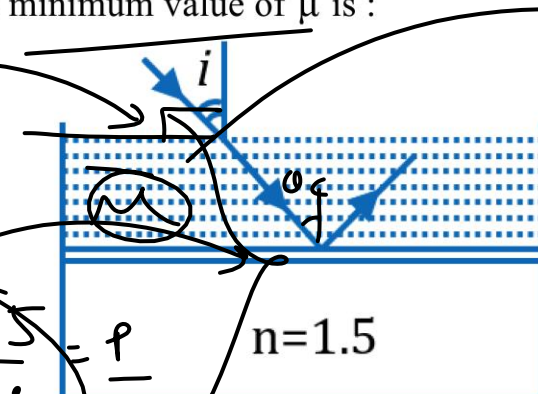
$$v = S(1 - e^{-\pi/2}) \Rightarrow t=2$$

$$W_{\text{all}} = \Delta KE \quad v = S(1 - e^{-1})$$

$$= \frac{1}{2} m (v^2 - u^2)$$

$$= \frac{1}{2} S^2 (1 - e^{-1})^2$$

4. Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ . A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is :



Handwritten notes and calculations:

- $\sin \theta_c = \frac{1}{\mu}$
- $\tan i_p = \mu_g = \frac{1.5}{\mu} = \frac{p}{B}$
- $\theta_c < i_p$
- $\sin \theta_c < \sin i_p$
- $\frac{1}{\mu} < \frac{1.5}{\sqrt{1.5^2 + \mu^2}}$
- $1.5^2 + \mu^2 < 1.5^2 \mu^2$
- $1.5^2 < (1.5^2 - 1)\mu^2$
- $\frac{2.25}{1.25} < \mu^2$
- $\sin i_p = \frac{1.5}{\sqrt{1.5^2 + \mu^2}}$

Options:

- (A) $\frac{3}{\sqrt{5}}$
- (B) $\frac{5}{\sqrt{3}}$
- (C) $\sqrt{\frac{5}{3}}$
- (D) $\frac{4}{3}$

$$\begin{array}{r} 459 \\ 2 \overline{) 225} \\ \underline{125} \\ 355 \end{array} < \mu^2$$

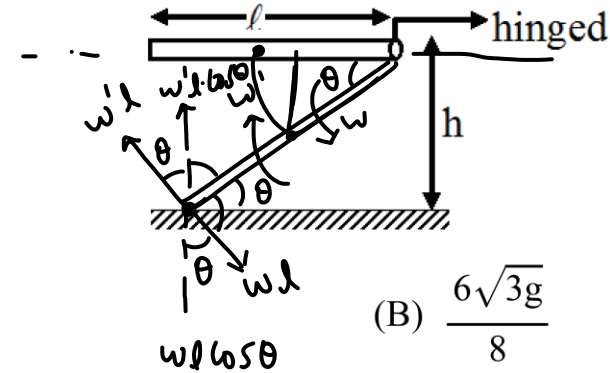
$$\frac{9}{5} < \mu^2$$

$$\mu^2 > \frac{9}{5}$$

$$\mu > \frac{3}{\sqrt{5}}$$

$$\mu_{\min} = \frac{3}{\sqrt{5}}$$

5. A thin rod of mass m and length ℓ is hinged at one end point which is at a distance h ($h < \ell$) above the horizontal surface. The rod is released from rest from the horizontal position. If e is the coefficient of restitution, the angular velocity of rod just after collision will be ($h = 1\text{m}$, $\ell = 2\text{m}$, $e = 1$) -



(A) $\frac{3\sqrt{3}g}{8}$

(B) $\frac{6\sqrt{3}g}{8}$

(C) $\frac{5\sqrt{3}g}{8}$

$\boxed{\omega' = \omega}$

(D) none of these

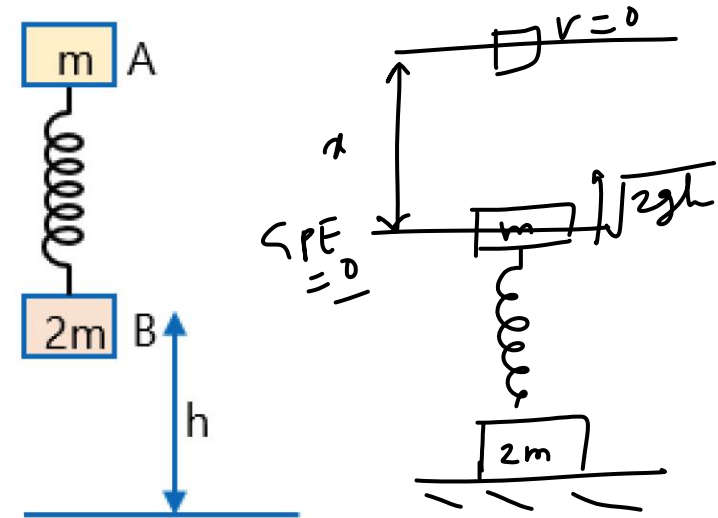
$mg \cdot \frac{h}{2} \cos \theta = \frac{1}{2} \cdot \frac{m\ell^2}{3} \times \omega^2$

$\cos \theta = \frac{1}{2}$
 $\sin \theta = \frac{\sqrt{3}}{2}$

$\frac{g}{2} = \frac{1^2}{2 \times 3} \times \omega^2$

$\omega = \frac{\sqrt{3g}}{2}$

6. From what minimum height h must the system be released when spring is unstretched so that after perfectly inelastic collision ($e = 0$) with ground, B may be lifted off the ground (Spring constant = k).



(A) $mg/(4k)$

(C) $mg/(2k)$

~~(B)~~ $4mg/k$

(D) none

$kx = 2mg$

$x = \frac{2mg}{k}$

$$\frac{1}{2} m (2\sqrt{gh})^2 = \frac{1}{2} k \left(\frac{2mg}{k} \right)^2 + mg \cdot \frac{2mg}{k}$$

$$h = \frac{2mg}{k} + \frac{2mg}{k}$$

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

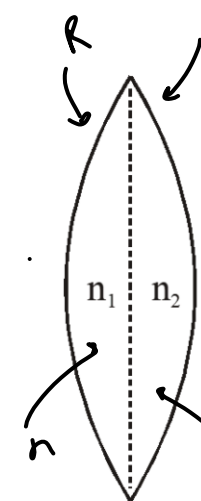
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A thin convex lens is made of two materials with refractive indices n_1 and n_2 , as shown in figure. The radius of curvature of the left and right spherical surfaces are equal. f is the focal length of the lens when $n_1 = n_2 = n$. The focal length is $f + \Delta f$ when $n_1 = n$ and $n_2 = n + \Delta n$. Assuming $\Delta n \ll (n-1)$ and $1 < n < 2$, the correct statement(s) is/are :



$$\frac{1}{f} = (n-1) \frac{2}{R}$$

$$f = \frac{R}{2(n-1)}$$

$$\frac{1}{f_{eq}} = (n-1) \left(\frac{1}{R} \right) + (n+\Delta n-1) \left(\frac{1}{R} \right)$$

$$= \frac{1}{R} [n-1 + n+\Delta n-1]$$

- (A) The relation between $\frac{\Delta f}{f}$ and $\frac{\Delta n}{n}$ remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.

(B) $\left| \frac{\Delta f}{f} \right| < \left| \frac{\Delta n}{n} \right|$

- (C) For $n = 1.5$, $\Delta n = 10^{-3}$ and $f = 20 \text{ cm}$, the value of $|\Delta f|$ will be 0.02 cm (round off to 2nd decimal place)

- (D) If $\frac{\Delta n}{n} < 0$ then $\frac{\Delta f}{f} > 0$

$$\frac{1}{f_{eq}} = \frac{1}{R} [2n-2+\Delta n]$$

$$f_{eq} = \frac{R}{2(n-1)+\Delta n}$$

$$f = \frac{R}{2(h-1)}$$

$$\Delta f = f_{\text{new}} - f$$

$$= \frac{R}{2(h-1) + \Delta h} - \frac{R}{2(h-1)}$$

$$\Delta f = R \left[\frac{2(h-1) - 2(h-1) - \Delta h}{[2(h-1)][2(h-1) + \Delta h]} \right]$$

$$\frac{\Delta f}{f} = \frac{-\Delta h}{2(h-1) + \Delta h} = \frac{-\Delta h}{2(h-1)} \times \frac{h}{h}$$

$$= -\frac{h}{2(h-1)} \left(\frac{\Delta h}{h} \right)$$

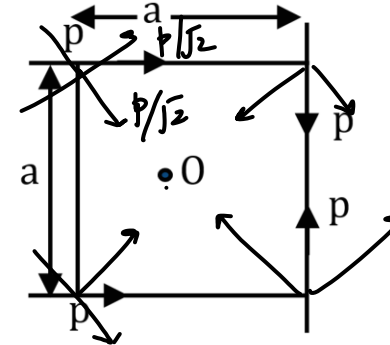
$$\left| \frac{\Delta f/f}{\Delta h/h} \right| = \oplus \frac{h}{2(h-1)}$$

$$= \frac{1}{2(1 - 1/n)}$$

$$\frac{\Delta f}{f} = -\frac{\Delta h}{2(h-1)} = -\frac{h}{2(h-1)} \frac{\Delta h}{h}$$

$$\left| \frac{\Delta f}{f} \right| > \left| \frac{\Delta h}{h} \right|$$

2. Four short dipoles each of dipole moment ' p ' are placed at the vertices of a square of side a . The direction of the dipole moments are shown in the figure.



(A) Electric field at O is $\frac{\sqrt{2}p}{2\pi\epsilon_0 a^3}$.

(B) Electric field at O is $\frac{\sqrt{2}p}{\pi\epsilon_0 a^3}$.

(C) Electrostatic potential at O is zero.

(D) Net dipole moment is $2p$.

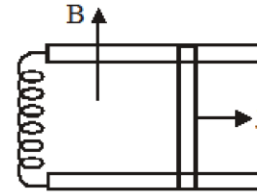
$$p_0 = p/\sqrt{2}$$

$$\frac{kp_0}{r^3} = E$$

$$E_{net} = 2\sqrt{2}E$$

$$= 2\sqrt{2} \times \frac{kp/\sqrt{2}}{(a/\sqrt{2})^3} = \frac{4\sqrt{2}kp}{a^3}$$

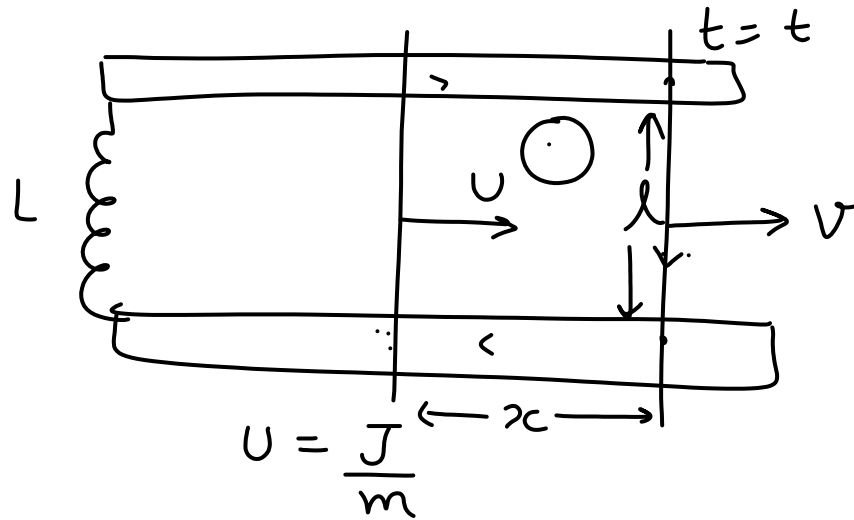
3. Two parallel resistanceless rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option (s).



- (A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor d

$$= \sqrt{\frac{3J^2 L}{4B^2 \ell^2 m}}$$
- (B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{4Lm}}$
- (C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor d

$$= \sqrt{\frac{3J^2 L}{B^2 \ell^2 m}}$$
- (D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{mL}}$



$$F = B i l$$

$$= B \left(\frac{B l x}{L} \right) l$$

$$F = \frac{B^2 l^2 x}{L}$$

$$\mathcal{E} = B v l = L \frac{di}{dt}$$

$$\int B v l dt = \int L di$$

$$B l \int v dt = L \int di$$

$$\boxed{B l x = L I}$$

$$\frac{B l x}{L} = I$$

$$V = \omega \sqrt{A^2 - x^2}$$

$$\frac{A\cancel{\omega}}{2} = \cancel{\omega} \sqrt{A^2 - x^2}$$

$$x = \pm \frac{\sqrt{3}}{2} A$$

$$x = \frac{\sqrt{3}}{2} \frac{J}{m\omega}$$

$$a = - \frac{B^2 l^2 x}{mL}$$

$$a = -\omega^2 x$$

S.H.M

$$\omega = \sqrt{\frac{B^2 l^2}{mL}} \star$$

$$U = \frac{J}{m} = A\omega$$

$$A = \frac{J}{m\omega}$$

C-II

(S)



$$f_R = \left(\frac{V \pm V_o}{V} \right) f_s$$

$$\frac{f_R}{f_s} = \left(\frac{V \pm V_o}{V} \right)$$

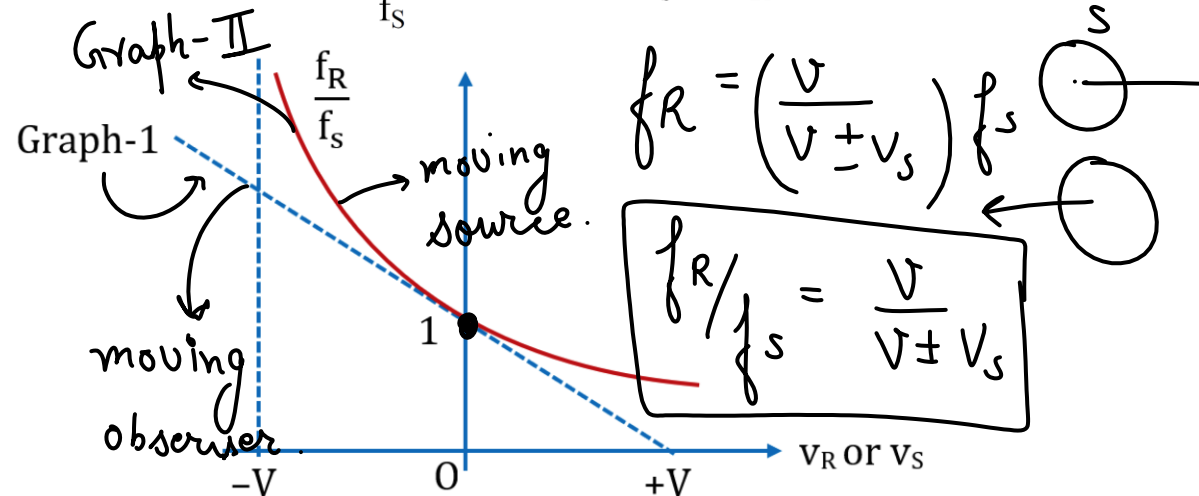
4. Consider :

Case-1 : A moving source moving with velocity v_s and a stationary receiver.

Case-2 : A moving receiver moving with velocity v_R and a stationary source.

In either case f_R represents frequency heard by receiver and f_s represents frequency emitted by source.

Figure shows graph of $\frac{f_R}{f_s}$ on y axis and v_s or v_R on x axis. Mark the **CORRECT** option (s)



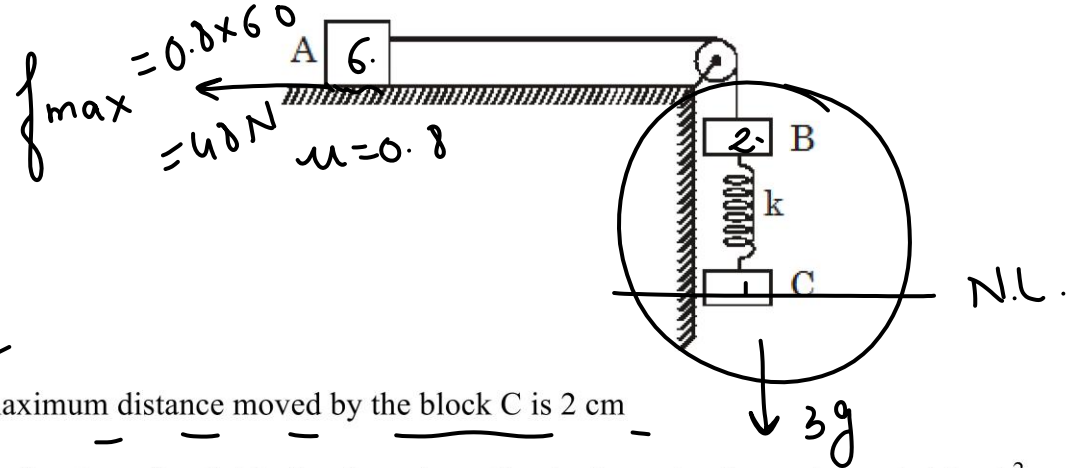
☒ (A) Graph-1 corresponds to moving receiver and Graph-2 corresponds to moving source.

☐ (B) Graph-2 corresponds to moving receiver and Graph-1 corresponds to moving source.

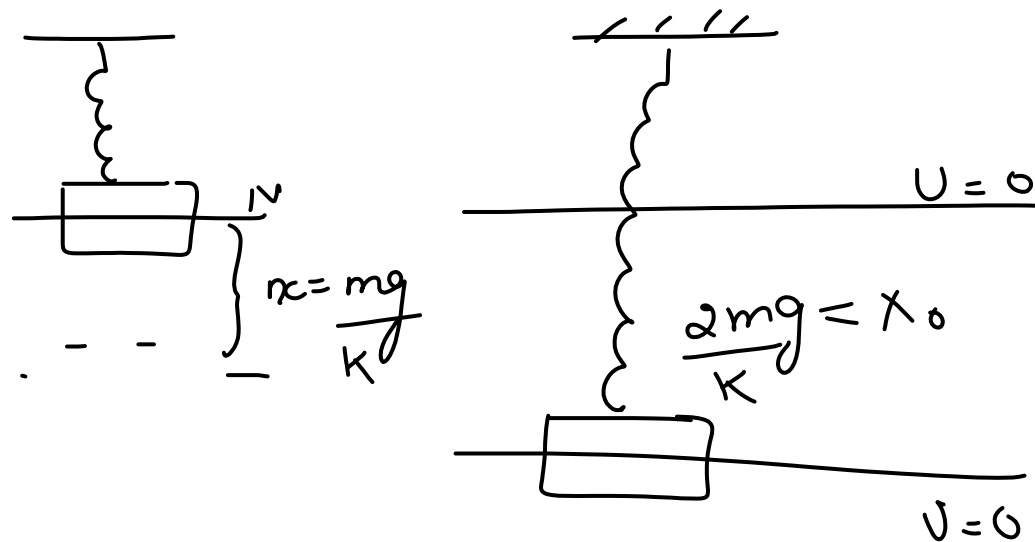
☒ (C) Graphs to left of origin represents approaching listener and to right represents receding listener

☒ (D) Graphs to right of origin represents approaching source and to left represent receding source

5. In the figure shown masses of the blocks A, B and C are 6 kg, 2 kg and 1 kg respectively. Mass of the spring is negligibly small and its stiffness is 1000 N/m. The co-efficient of friction between the block A and the table is $\mu = 0.8$. Initially block C is held such that spring is in relaxed position. The block is released from rest. Choose the correct statement (s) :-



- (A) The maximum distance moved by the block C is 2 cm
- ~~(B) the acceleration of each block, when elongation in the spring is maximum is 10 m/s^2~~
- (C) If the block C oscillates about its equilibrium position with amplitude $\frac{m_c g}{2k}$, the value of the tension in the string as a function of time is given by $T = m_B g + m_C g + kA \sin \omega t$ (where $A = \frac{m_c g}{2k}$, $\omega = \sqrt{\frac{k}{m_C}}$ (At the time $t = 0$ the block C is in equilibrium and is moving downward.)
- (D) If the block C oscillates about its equilibrium position with amplitude $\frac{m_c g}{2k}$, the value of the tension in the string as a function of time is given by $T = kA \sin \omega t$ (where $A = \frac{m_c g}{2k}$, $\omega = \sqrt{\frac{k}{m_C}}$ (At the time $t = 0$ the block C is in equilibrium and is moving downward.)

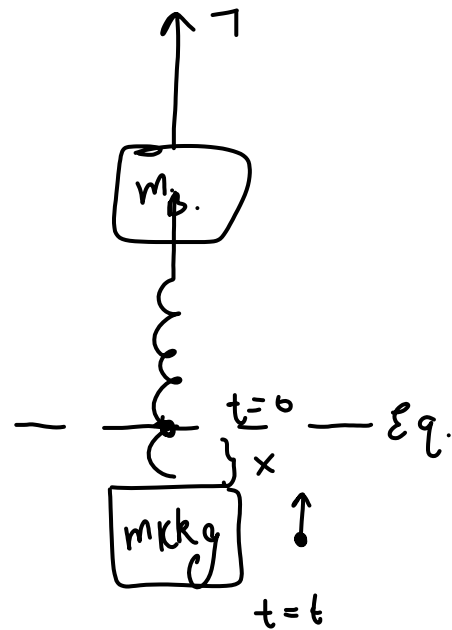


$$\frac{1}{2} K x^2 = mg x$$

$$x = \frac{2mg}{K}$$

$$X_{max} = \frac{2 \times 1 \times 10}{1000} = 2 \text{ cm}$$

$$a = \omega^2 x$$



$$\sum F_{\text{net}} = \sum m_i a_i$$

$$T - m_B g - m_C g = m_C a$$

$$= m_C \omega^2 x$$

$$= \boxed{m_C \omega^2} A \sin \omega t$$

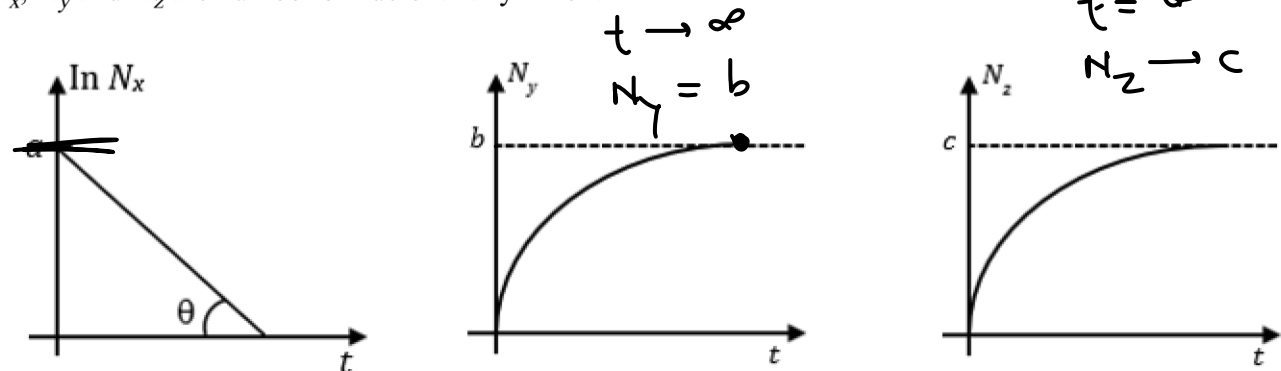
$$T = m_B g + m_C g + K A \sin \omega t$$

6. A sample of unstable radioactive nuclei X can decay into two stable nuclei Y and Z .



At $t = 0$ there are no nuclei of Y and Z . Three graphs $\ln N_x$ versus t , N_y versus t and N_z versus t are drawn as shown in figure.

N_x , N_y and N_z are number of nuclei at any time t .



- (A) Ratio of $N_y : N_z$ at any time is constant.
- (B) Decay constant for decay of X into Y is $\frac{b \tan \theta}{b + c}$.
- (C) Decay constant for decay of X into Z is $\frac{c \tan \theta}{e^a}$.
- (D) Ratio of $N_y : N_z$ is time dependent.

$$\ln \left(\frac{N_{xc}}{N_0} \right) = -(\lambda_1 + \lambda_2)t$$

$$N_{xc} = N_0 e^{-(\lambda_1 + \lambda_2)t}$$

$$-\frac{dN_{xc}}{dt} = \frac{dN_y}{dt} + \frac{dN_z}{dt}$$

$$-\frac{dN_{xc}}{dt} = \lambda_1 N_{xc} + \lambda_2 N_{xc}$$

$$\int_{N_0}^{N_{xc}} \frac{dN_{xc}}{N_{xc}} = \int_0^t -(\lambda_1 + \lambda_2) dt$$

$$\star \ln N_{xc} - \ln N_0 = -(\lambda_1 + \lambda_2)t$$

$$y - c = -mx$$

$$\ln N_0 = a$$

$$N_0 = e^a$$

$$\tan \theta = \lambda_1 + \lambda_2 \quad \star$$

$$\frac{dN_y}{dt} = \lambda_1 N_x$$

$$\frac{dN_y}{dt} = \lambda_1 N_0 e^{-(\lambda_1 + \lambda_2)t}$$

$$N_y \int_0^t dN_y = \lambda_1 N_0 \int_0^t e^{-(\lambda_1 + \lambda_2)t} dt$$

$$N_y = \frac{\lambda_1 N_0 \left[e^{-(\lambda_1 + \lambda_2)t} \right]_0^t}{-(\lambda_1 + \lambda_2)} = \frac{\lambda_1 N_0 (e^{-(\lambda_1 + \lambda_2)t} - 1)}{-(\lambda_1 + \lambda_2)}$$

$$N_y = \frac{\lambda_1 N_0}{\lambda_1 + \lambda_2} \left[1 - e^{-(\lambda_1 + \lambda_2)t} \right]$$

$$N_z = \frac{\lambda_2 N_0}{\lambda_1 + \lambda_2} \left[1 - e^{-(\lambda_1 + \lambda_2)t} \right]$$

$$\frac{N_y}{N_z} = \frac{\lambda_1}{\lambda_2} = \frac{b}{c}$$

$$\lambda_1 = \frac{b}{c} \lambda_2$$

$$\lambda_1 = \frac{b}{c} (\tan \theta - \lambda_1)$$

$$\lambda_1 \left(1 + \frac{b}{c} \right) = \frac{b}{c} \tan \theta$$

$$\lambda_1 (b+c) = b \tan \theta$$

$$\lambda_1 = \frac{b \tan \theta}{b+c}$$

$$b = \frac{\lambda_1 N_0}{\lambda_1 + \lambda_2}$$

$$c = \frac{\lambda_2 N_0}{\lambda_1 + \lambda_2}$$

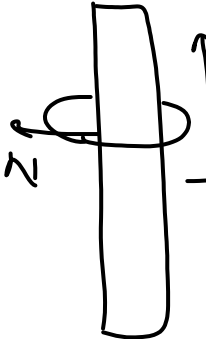
7. A small charged bead of charge $+Q$ mass m is threaded on a long smooth rod oriented along y axis. An electromagnetic wave $\vec{E} = E_0 \hat{j} \sin(kx - \omega t)$ is incident on the system. Choose the correct statement (s). Neglect gravity. Initially bead was at rest.

(A) The maximum speed of the bead is $\frac{QE_0}{m\omega}$.

~~(B)~~ The maximum speed of the bead is $\frac{2QE_0}{m\omega}$.

~~(C)~~ The average force on the bead over one time period due to magnetic field of the wave is zero.

(D) If the bead was negatively charged, the average force on the bead over one time period due to magnetic field is in positive x direction.

$\hat{j} \times \hat{k} = \hat{i}$

 $\vec{B} = \frac{E_0}{c} \sin(kx - \omega t) \hat{k}$
 $\Rightarrow \int m dv = - \int q E_0 \sin(kx - \omega t) dx$
 $\Rightarrow m v = \frac{q E_0}{\omega} (\cos \omega t)_t$

$$v = \frac{qE_0}{m\omega} (\cos \omega t - 1) \hat{j}$$

$$\Rightarrow |v| = \left| \frac{qE_0}{m\omega} \underbrace{(1 - \cos \omega t)}_{\text{max}} \right|$$

$$\Rightarrow v_{\text{max}} = \frac{2qE_0}{m\omega}$$

$$\begin{aligned} \vec{F}_B &= q \vec{v} \times \vec{B} = q \left[\frac{qE_0}{m\omega} (\cos \omega t - 1) \hat{j} \right] \times \left[\frac{B_0}{c} \sin(-\omega t) \hat{k} \right] \\ &= -\frac{q^2 E_0^2}{m\omega c} (\cos \omega t - 1) (\sin \omega t) \hat{i} \end{aligned}$$

$$|\vec{F}_B| = \frac{q^2 E_0^2}{m\omega c} \left(\cancel{\sin \omega t}^0 - \cancel{\sin \omega t \cos \omega t}^0 \right) \hat{i}$$

$$\boxed{F_{\text{avg}} = 0}$$

8. A ball of mass m has an acceleration $a = v^2$ along X-axis. At $t = 0$, $x = 0$ and $v = v_0$.

~~(A)~~ speed of ball as a function of position is $v = v_0 e^x$

(B) position of the ball as a function of time is $x = \ln(1 - v_0 t)$

(C) acceleration of body as a function of position is $a = v_0 e^x$

~~(D)~~ speed of ball as a function of time is $\frac{V_0}{1 - V_0 t}$

$$\begin{aligned}
 & \underline{a = v^2} \\
 \Rightarrow & v \frac{dv}{dx} = v^2 \\
 \Rightarrow & \int_{v_0}^v \frac{dv}{v} = \int_0^x dx \\
 & \ln \left(\frac{v}{v_0} \right) = x \Rightarrow \boxed{v = v_0 e^x} \quad \times \\
 & \Downarrow \\
 & \boxed{a = (v_0 e^x)^2} \\
 & \frac{dx}{dt} = v_0 e^x \\
 \Rightarrow & \int_0^x e^{-x} dx = \int_0^t v_0 dt \\
 \Rightarrow & -\left(e^{-x}\right)_0^x = v_0 t
 \end{aligned}$$

$$\Rightarrow (e^{-u} - 1) = -V_0 t$$

$$\Rightarrow e^{-u} = 1 - V_0 t$$

$$\Rightarrow -u = \ln(1 - V_0 t) \Rightarrow \boxed{u = -\ln(1 - V_0 t)}$$

$$\frac{du}{dt} = v = -\frac{1}{(1 - V_0 t)} (-V_0)$$

$$\left\{ v = \frac{V_0}{1 - V_0 t} \right\}$$

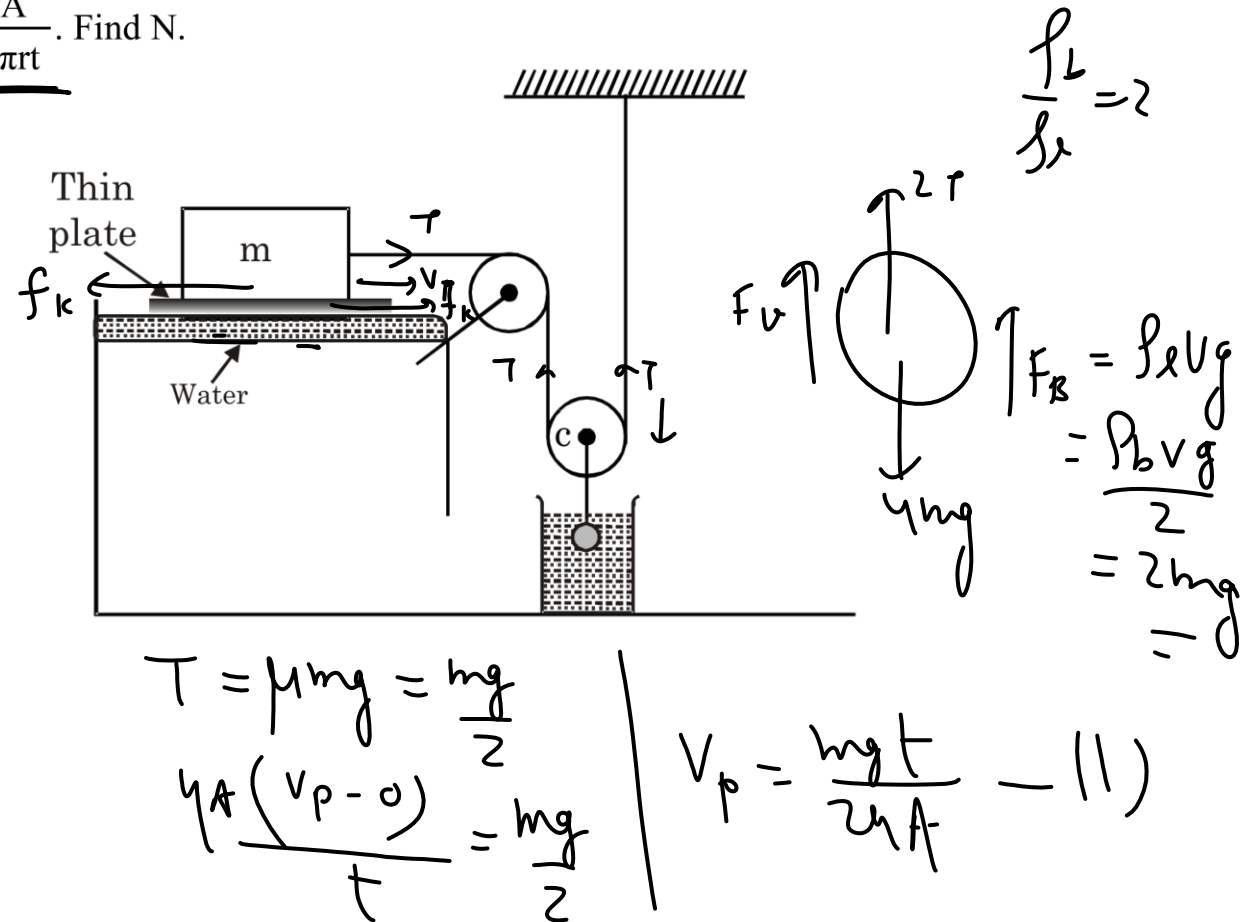
SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33 , -0.30 , 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A thin water layer of thickness t is spread over a horizontal surface. A thin plate of mass m and cross-sectional area A is placed over it. A block of mass m is kept over the plate which is connected to a spherical body of mass $4m$, radius r and specific gravity 2 as shown in the figure. Block is sliding over the plate. Cof between the plate and the block is 0.5 . The ratio of terminal velocities attained by the sphere and plate is $\frac{2A}{N\pi r t}$. Find N .



$$\Rightarrow mg + 2mg + 6\pi\eta r v_s = 4mg$$

$$\Rightarrow v_s = \frac{mg}{6\pi\eta r} \quad \text{--- (11)}$$

$$\begin{aligned} \Rightarrow \frac{v_s}{v_p} &= \frac{\cancel{mg}}{6\pi\eta r} \times \frac{24\eta A}{\cancel{mgt}} \\ &= \frac{2A}{6\pi r t}, \quad \underline{\underline{N=6}} \end{aligned}$$

2. A satellite is orbiting around the earth in a circular orbit. Its orbital speed is V_0 . A rocket on board is fired from the satellite which imparts a thrust to the satellite directed radially away from the centre of the earth. The duration of the engine burn is negligible so that it can be considered instantaneous. Due to this thrust a velocity variation ΔV is imparted to the satellite. Find the minimum value of the ratio $\frac{\Delta V}{V_0}$ for which the satellite will escape out of the gravitational field of the earth.

$$\frac{1}{2} m v^2 - \frac{G M m}{r} = 0$$

$$(V_0^2 + \Delta V^2) = 2V_0^2$$

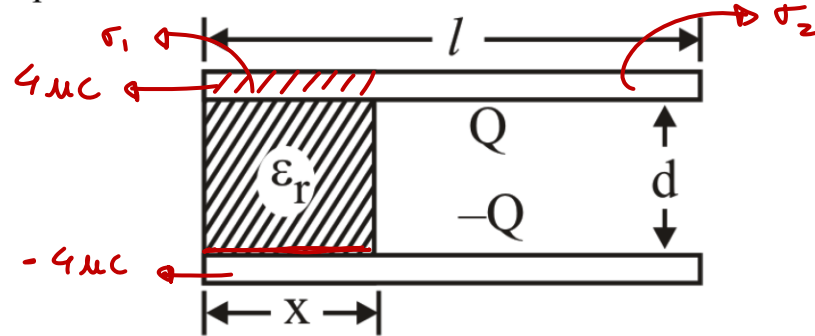
$$\Rightarrow V_0 = \Delta V$$

$$\Rightarrow \frac{\Delta V}{V_0} = 1$$

$$V_0 = \sqrt{\frac{GM}{r}}$$

$$V = \sqrt{V_0^2 + \Delta V^2}$$

3. Two conducting square plates with length l are arranged parallel to each other at a distance d ($\ll l$). The space between the plates is filled with a dielectric of relative permittivity $\epsilon_r = 4$ up to a length $x = l/3$. The upper plate is given a charge Q and the lower plate is given a charge $-Q$ as shown in the figure. If $Q = 6\mu\text{C}$ then find how much bound charge (in μC) is present on the lower surface of dielectric?



$$Q' = \frac{2Q}{l^2} \left(\frac{l^2}{3} \right)$$

$$Q' = \frac{2(6)}{3}$$

$$E_1 = E_2$$

$$\frac{\sigma_1}{4\epsilon_0} = \frac{\sigma_2}{\epsilon_0}$$

$$\sigma_1 = 4\sigma_2$$

$$\sigma_1 = \frac{2Q}{l^2}$$

$$\sigma_1(xl) + \sigma_2(l-x)l = Q$$

$$4\sigma_2 \left(\frac{l^2}{3} \right) + \sigma_2 \left(\frac{2l^2}{3} \right) = Q$$

$$\sigma_2 = \frac{Q}{2l^2}$$

$$Q_b = 4 \left(1 - \frac{1}{4} \right) = 3\mu\text{C} \text{ Ans.}$$

4. A solid has a volume v_0 when external pressure is p_0 . Bulk modulus of the material is B . Now the pressure is increased to p . The additional energy per unit volume now stored in the material is $\frac{3(p^2 - p_0^2)}{zB}$. Find $z = ?$

$$dp = -B \frac{dv}{v} \Rightarrow dv = -\frac{Vdp}{B}$$

$$W = \int p dv$$

$$= \int p \left(-\frac{V}{B} dp\right)$$

$$W = -\frac{V}{B} \int p dp$$

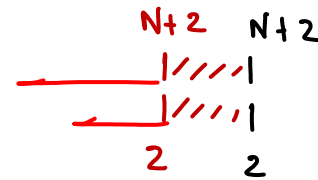
$$E = \frac{V}{B} \int_{p_0}^p p dp$$

$$\frac{E}{V} = \frac{1}{B} \left[\frac{p^2 - p_0^2}{2} \right]$$

$$\frac{3}{z} = \frac{1}{2}$$

$$z = 6 \text{ Ans.}$$

5. In a vernier calipers the main scale and the vernier scale are made up of different materials. When the room temperature increases by $\Delta T^\circ\text{C}$, it is found the reading of the instrument remains the same. Earlier it was observed that the front edge of the wooden rod placed for measurement crossed the N^{th} main scale division and $(N + 2)$ MSD coincided with the 2^{nd} VSD. Initially, 10 VSD coincided with 9 MSD. If coefficient of linear expansion of the main scale is α_1 and that of the vernier scale is α_2 . If α_1/α_2 is $\frac{x + 0.8}{(N + 2)}$ is (Ignore the expansion of the rod on heating)



$$10 \text{ VSD} = 9 \text{ MSD}$$

$$\text{VSD} = 0.9 \text{ MSD}$$

$$(N+2) \text{ MSD } \alpha_1 \Delta T = 2 \text{ VSD } \alpha_2 \Delta T$$

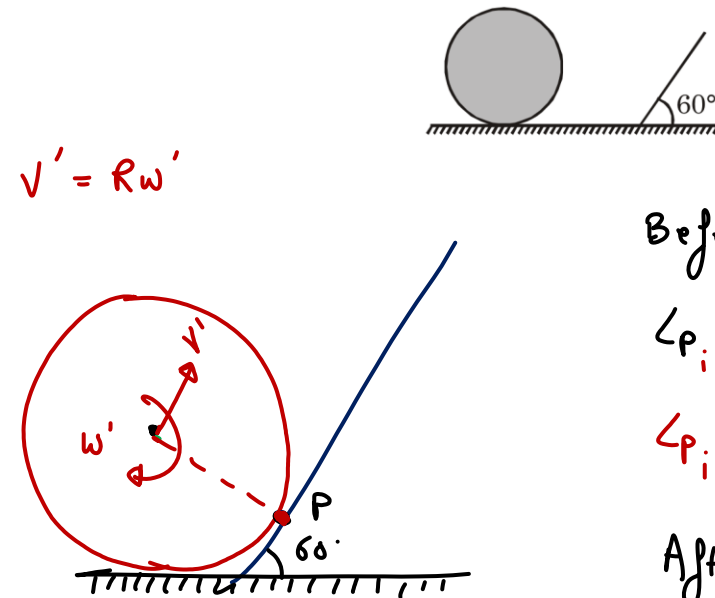
$$\frac{\alpha_1}{\alpha_2} = \frac{2 \text{ VSD}}{(N+2) \text{ MSD}}$$

$$= \frac{1.8}{N+2}$$

$$x + 0.8 = 1.8$$

$$x = 1 \text{ Ans.}$$

6. A uniform solid sphere rolling without slipping along a horizontal plane with angular speed $\omega_0 = 14$ rad/sec, suddenly collides with a plane inclined at an angle $\theta = 60^\circ$ as shown in the figure. Find the angular velocity (in rad/sec) of sphere just after impact : (assume that there is no slipping anytime and collision is perfectly inelastic)



$$V_0 = R\omega_0$$

$$7 + \frac{28}{5} = \frac{7}{5} \omega'$$

$$\omega' = 9 \text{ rad/s.}$$

Before collision

$$L_{P_i} = mV_0 \frac{R}{2} + I_c \omega_0$$

$$L_{P_i} = \frac{mR^2}{2} \omega_0 + \frac{2}{5} mR^2 \omega_0$$

After collision

$$L_{P_f} = mV'R + \frac{2}{5} mR^2 \omega'$$

$$= mR^2 \omega' + \frac{2}{5} mR^2 \omega'$$

$$= \frac{7}{5} mR^2 \omega'$$

JEE

ADVANCED 2024



FINAL PRACTICE
PAPER-1

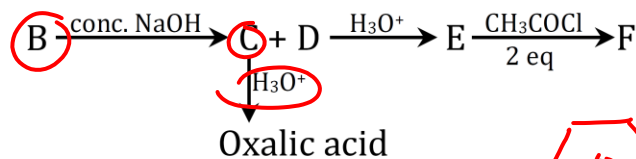


CHEMISTRY

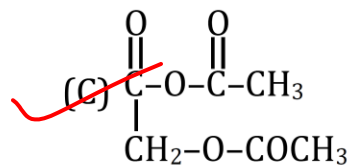
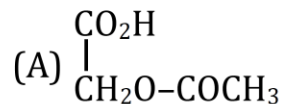
SECTION-I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

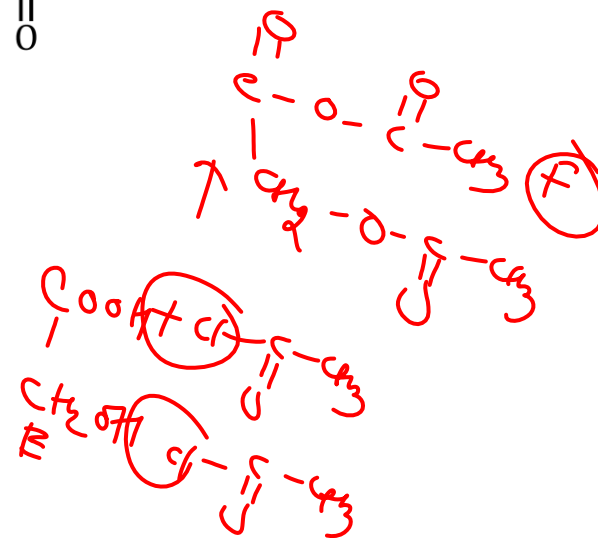
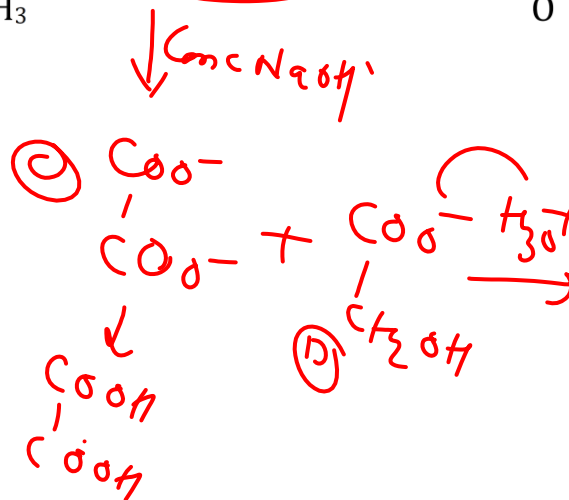
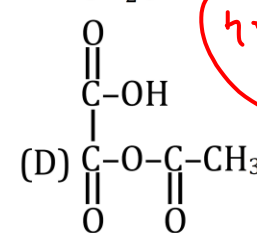
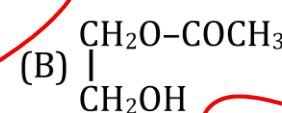
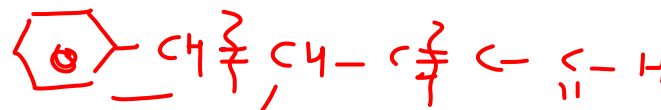
1. An aldehyde A ($C_{11}H_{18}O$) which does not undergo self aldol condensation, gives benzaldehyde and two moles of B on ozonolysis. Compound B which can reduce ammoniacal silver nitrate and itself oxidizes to oxalate salt.



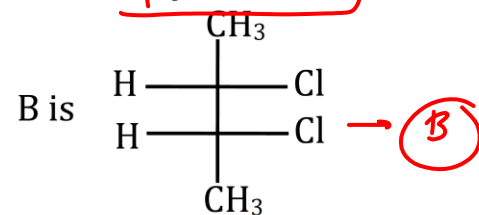
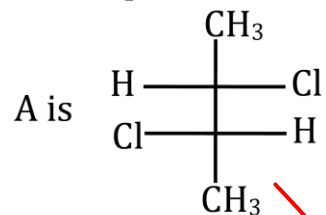
F is



$$A \rightarrow C_{11}H_{18}O \quad D.O = (11+1) - \left(\frac{18}{2}\right) = 8 - 3$$



2. If the specific rotation produced by A is $+36^\circ$, then the specific rotation produced by B will be



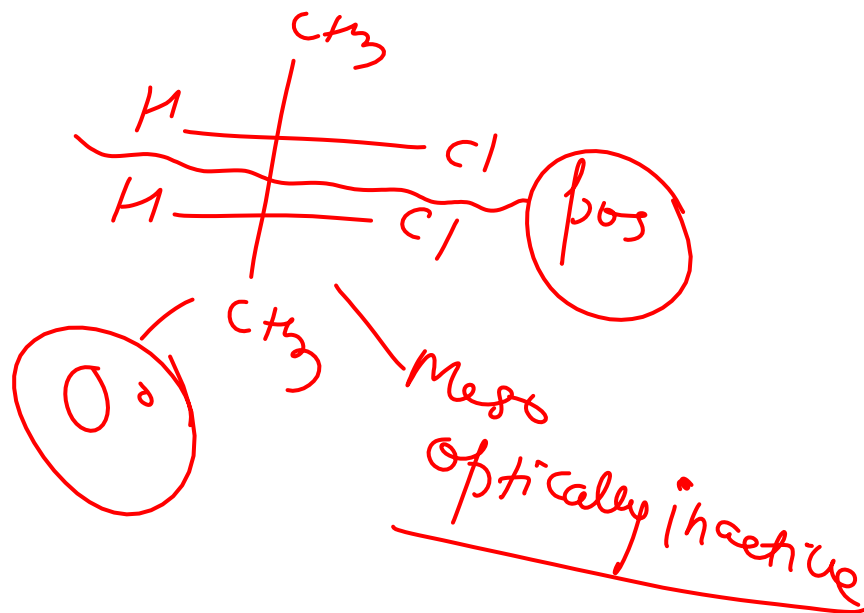
(A) -36°

(C) 0°

$+36^\circ$

(B) $+36^\circ$

(D) it is experimental, can't be predicted



Paragraph.

	Column-I		Column-I
(I)		(P)	
(II)		(Q)	Recemic mixture
(III)		(R)	Meso compound
(IV)		(S)	CO ₂ gas is evolved

3.

The correct combination in the following is

(A) I, P, S

(B) I, Q, S

(C) II, Q, R

(D) II, P, R

4. For the following reaction : $2A + B + C \rightarrow 3D$, rate was measured at different concentrations and noted. Identify the option which is correct on the basis of data.

Exp.	conc. of [A]	conc. of [B]	conc. of [C]	Rate of formation of [D]
(1)	0.03 M	0.04 M	0.02 M	$10^{-3} \text{ M min}^{-1}$
(2)	0.06 M	0.08 M	0.02 M	$10^{-3} \text{ M min}^{-1}$
(3)	0.06 M	0.16 M	0.04 M	$5 \times 10^{-4} \text{ M min}^{-1}$
(4)	0.06 M	0.08 M	0.04 M	$10^{-3} \text{ M min}^{-1}$

- ~~(A)~~ Overall order of the reaction is negative
~~(B)~~ Order of the reaction with respect to A will be 1.
~~(C)~~ Overall order of the reaction is 2
~~(D)~~ Order of the reaction with respect to B will be 1.

$$\left. \begin{array}{l} x = 1 \\ y = -1 \\ z = 0 \end{array} \right\}$$

$$r = k[A]^x[B]^y[C]^z$$

(B) ~~A~~

$$10^{-3} = K [0.03]^x [0.04]^y [0.02]^z \quad \text{--- (1)}$$

$$10^{-3} = K [0.06]^x [0.08]^y [0.02]^z \quad \text{--- (2)}$$

$$5 \times 10^{-4} = K [0.06]^x [0.16]^y [0.04]^z \quad \text{--- (3)}$$

$$10^{-3} = K [0.06]^x [0.08]^y [0.04]^z \quad \text{--- (4)}$$

$$\text{eq. (3)} \div \text{eq. (4)}$$

$$0.5 = (2)^y$$

$$(2)^{-1} = (2)^y$$

$$y = -1$$

$$\text{eq. (1)} \div \text{eq. (2)}$$

$$1 = \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^z$$

$$x = 1$$

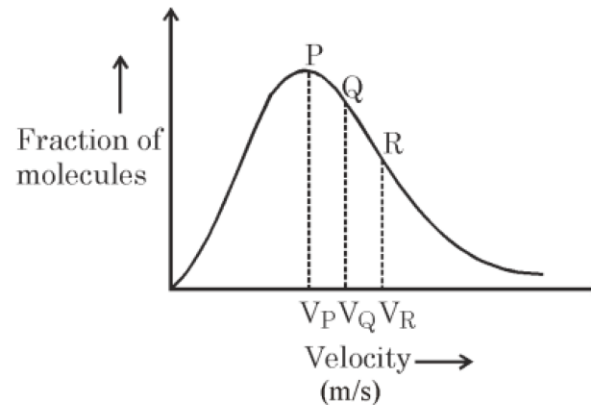
$$\text{eq. (2)} \div \text{eq. (3)}$$

$$2 = \left(\frac{1}{2}\right)^y \left(\frac{1}{2}\right)^z$$

$$z = 0$$

$$M = 4 \text{ gm} = 4 \times 10^{-3} \text{ kg}$$

5. Following maxwell curve is plotted for helium gas at $\frac{2000}{8.314}$ K temperature.



$$V_P = \sqrt{\frac{2RT}{m}} \quad \cancel{RT = \frac{2000}{8.314}}$$

$$V_Q = \sqrt{\frac{8RT}{\pi m}} \quad RT = 2000$$

$$V_R = \sqrt{\frac{3RT}{m}}$$

Calculate the value of Z (in S.I. units) ? $[\pi = 3.14]$

$$\text{Where } Z = \left[3.14 V_Q^2 + 2(V_P^2 - V_R^2) \right] \times 10^{-7}$$

(A) 0.6

(B) 0.3

(C) 0.05

(D) 0.15

$$Z = \left\{ \cancel{3.14 \left(\frac{8RT}{\pi m} \right)} + 2 \left(\frac{2RT}{m} - \frac{3RT}{m} \right) \right\} \times 10^{-7}$$

$$Z = \left[\frac{8 \times 2000}{4 \times 10^{-3}} + 2 \left(\frac{2 \times 2000}{4 \times 10^{-3}} - \frac{3 \times 2000}{4 \times 10^{-3}} \right) \right] \times 10^{-7}$$

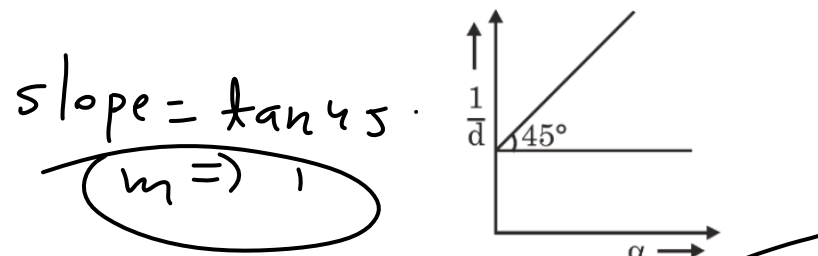
$$\left[4000 + (2000 - 3000) \right] \frac{10^{-7}}{10^{-3}}$$

$$\Rightarrow 3000 \times 10^{-4}$$

$$\Rightarrow 0.3$$

$$(B) \quad 0.3 \quad \underline{\underline{A_2}}$$

6. For the equilibrium $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ of reaction starts with pure NH_3 . Density $d(\text{gm/Lit.})$ of equilibrium mixture varies with degree of dissociation (α) of NH_3 of 237°C in such a manner that $\frac{1}{d}$ v/s " α " is a linear plot shown below, what is the total pressure at equilibrium.



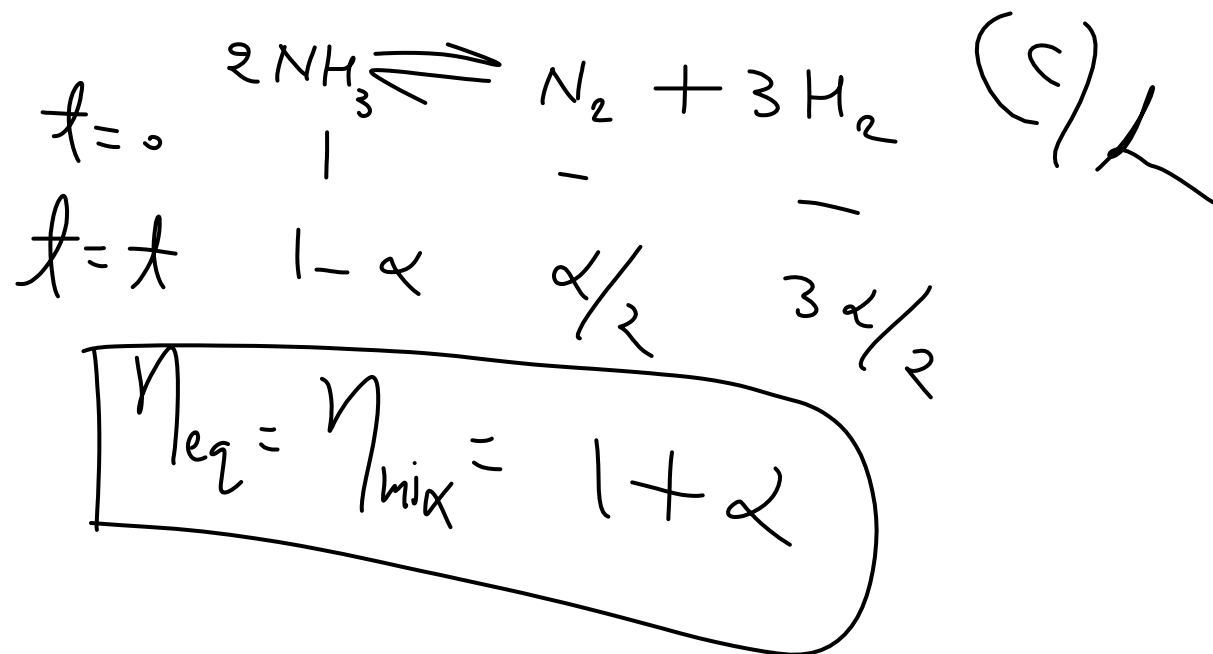
$T = 237^\circ\text{C}$
 $\Rightarrow 237 + 273$
 $\Rightarrow 510\text{ K}$

(A) 5.110 atm

(B) 3.862 atm

(C) 2.463 atm

(D) 1.56 atm





$$M < \frac{1}{h}$$

$$\frac{M_{\text{mix}}}{M_{\text{int}}} = \frac{n_{\text{int}}}{n_{\text{mix}}}$$

$$\Rightarrow \frac{M_{\text{mix}}}{17} = \frac{1}{(1+\alpha)}$$

$$P M_{\text{mix}} = d_{\text{mix}} R T$$

$$P \left(\frac{17}{1+\alpha} \right) = d_{\text{mix}} R T$$

$$\frac{1}{d} = \frac{R T}{P(17)} (1+\alpha)$$

$$\frac{1}{d} = \frac{R T}{P(17)} + \frac{R T}{P(17)} (\alpha)$$

$$y = c + m x$$

$$\text{slope} = 1$$

$$\frac{R T}{P(17)} = 1$$

$$P = \frac{R T}{17}$$

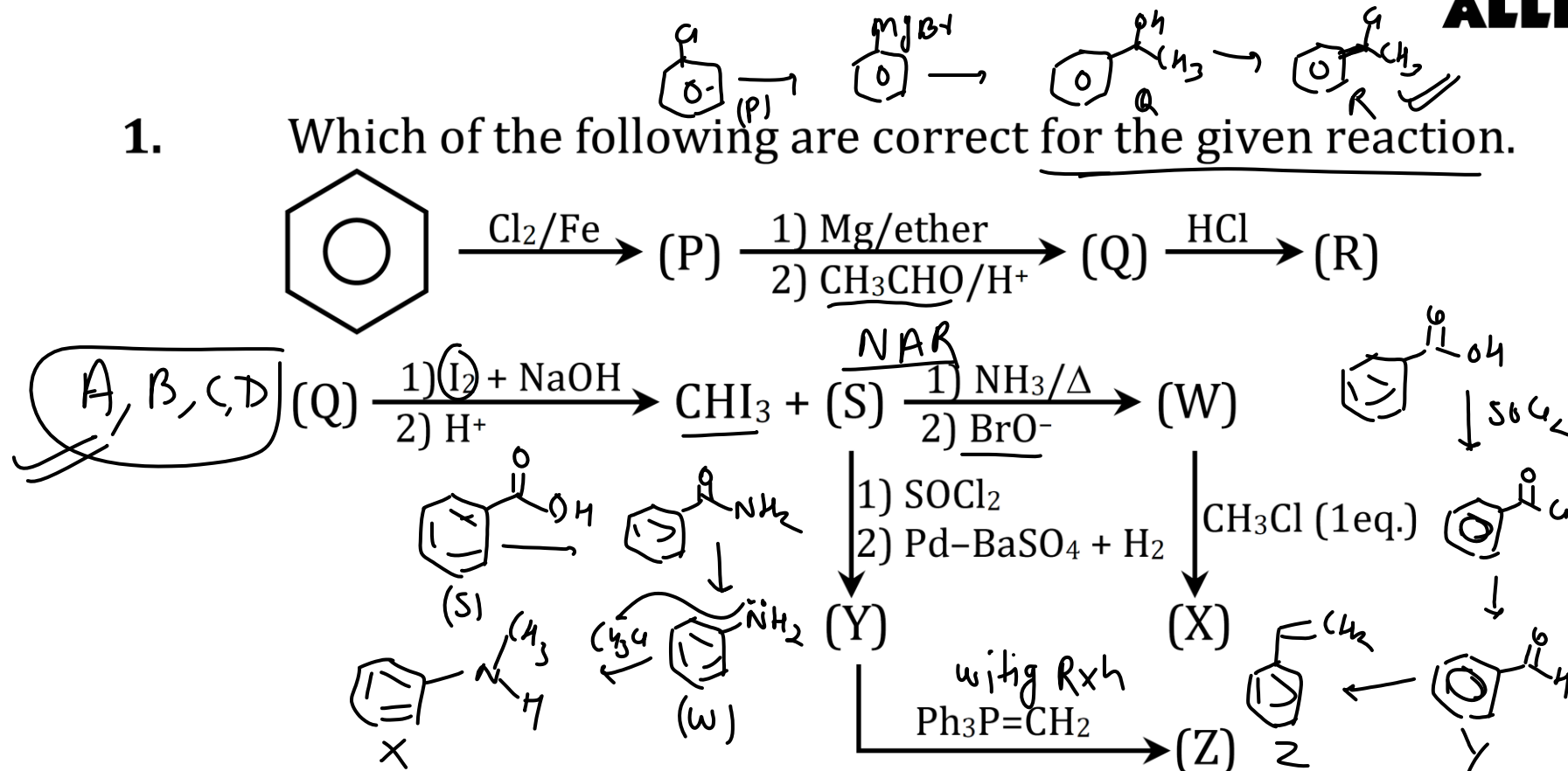
$$\Rightarrow \frac{0.0821 \times 30}{17}$$

$$\Rightarrow 2.463 \text{ atm}$$

SECTION-II : (Maximum Marks : 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 If only (all) the correct option(s) is(are) chosen;
 - Partial Marks* : +3 If all the four options are correct but ONLY three options are chosen;
 - Partial Marks* : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
 - Partial Marks* : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
 - Zero Marks* : 0 If unanswered;
 - Negative Marks* : -2 In all other cases.
- For example, If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

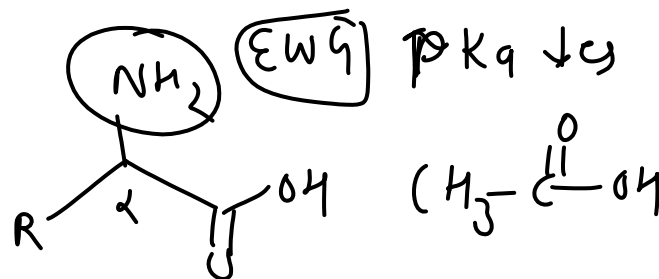
1. Which of the following are correct for the given reaction.



- (A) 'P' and 'R' can be distinguished by AgNO₃
- (B) 'Q' and 'S' can be distinguished by NaHCO₃
- (C) 'W' and 'X' can be distinguished by CHCl₃ + KOH
- (D) 'Y' and 'Z' can be distinguished by Tollen's reagent

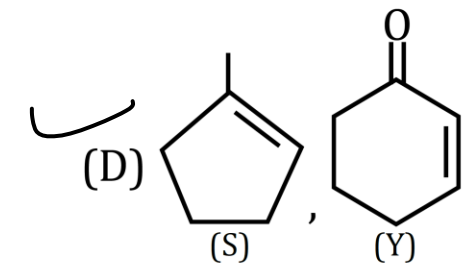
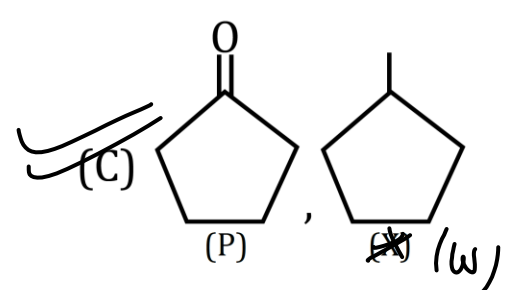
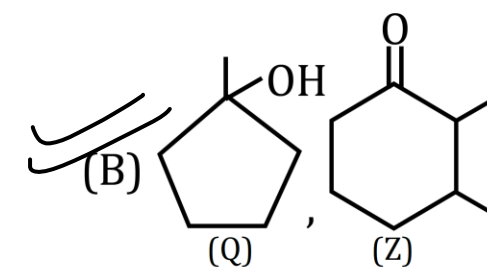
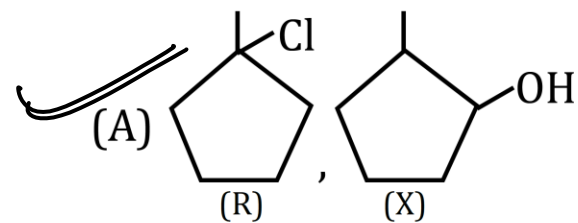
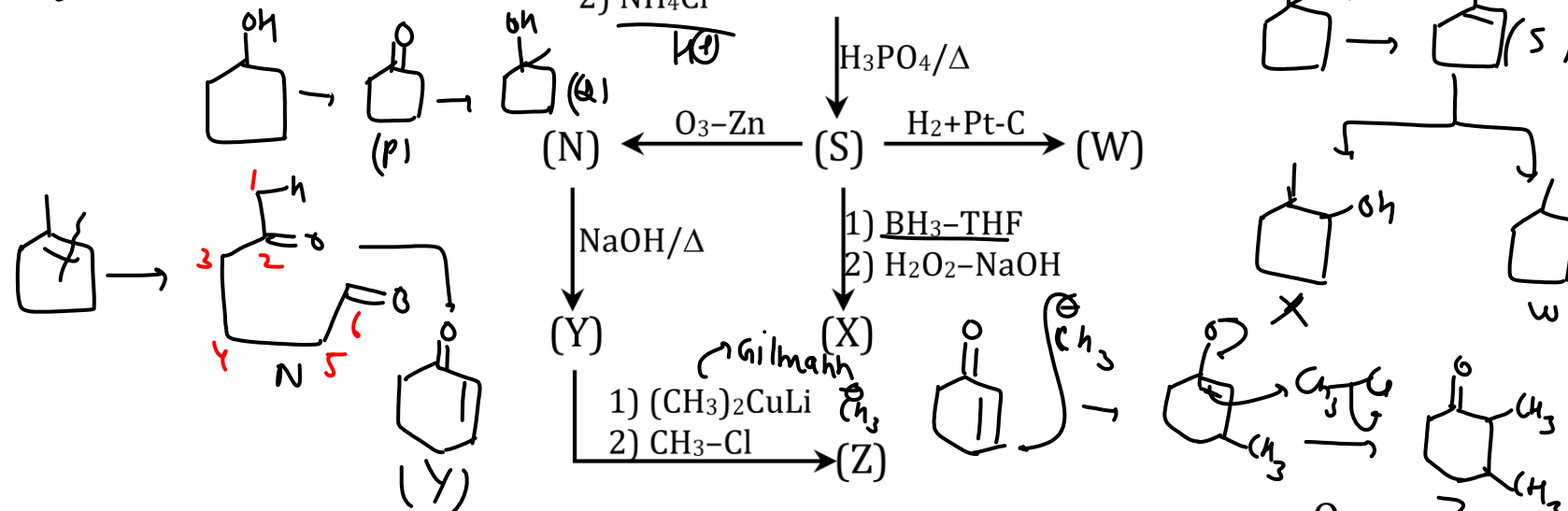
2. Correct statement(s) among the following is/are

- ☒ (A) The pK_a of $-COOH$ group in an α -amino acid is lower than the pK_a of acetic acid.
☒ (B) Proline is an example of aromatic heterocyclic ring containing amino acids
☒ (C) The pentaacetate of glucose does not react with hydroxylamine
☒ (D) α -D - (+) - Glucopyranose and β -D - (+) - Glucopyranose are differ in their configuration at C_2 position



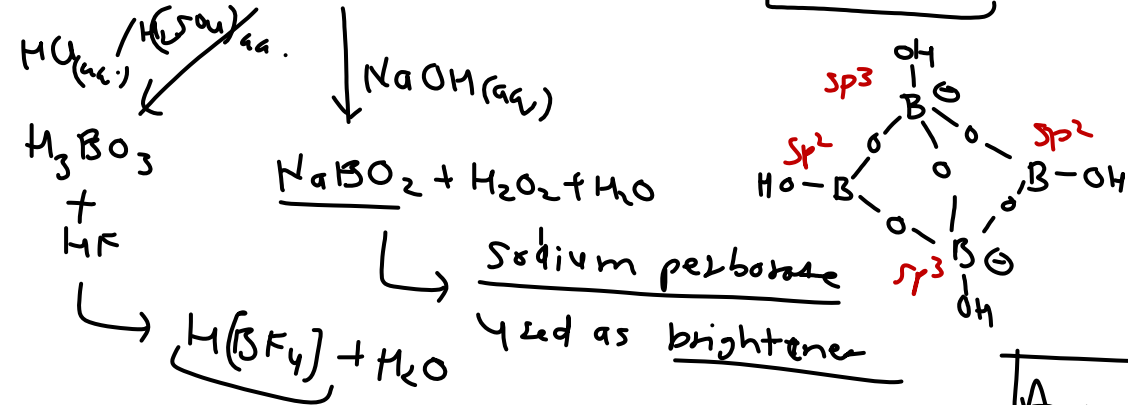
A, C

$$\text{D. } \text{C}_5\text{H}_{10}\text{O} \xrightarrow{\text{PCC}} (\text{P}) \xrightarrow[\text{2) NH}_4\text{Cl}]{\text{1) CH}_3\text{MgCl}} (\text{Q}) \xrightarrow{\text{HCl}} (\text{R})$$

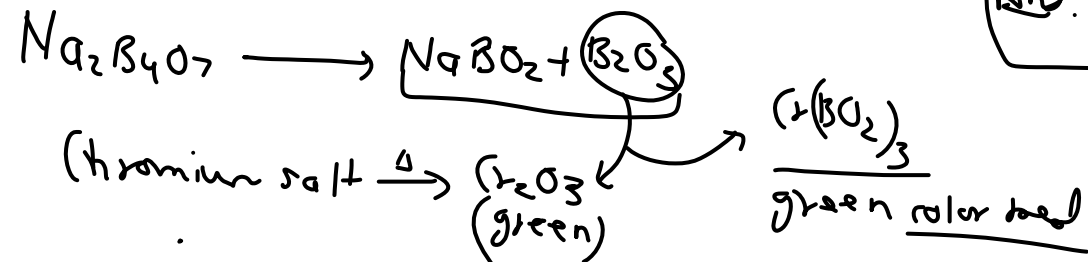


4. Colemanite $\text{Ca}_2\text{B}_6\text{O}_{11} + \text{Na}_2\text{CO}_3 \xrightarrow{\Delta} [\text{X}] + \text{CaCO}_3 + \text{NaBO}_2$ (Unbalanced equation) Correct choice for [X] is /are
- (A) ~~Structure of anion of crystalline [X] has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised.~~
- (B) [X] with NaOH (aq.) gives a compound which on reaction with hydrogen peroxide in alkaline medium yields a compound used as brightness in soaps.
- (C) ~~Hydrolysis of [X] with HCl or H₂SO₄ yields a compound which on reaction with HF gives fluoroboric acid~~
- (D) ~~[X] on heating with chromium salt in oxidizing flame gives green coloured bead~~

(A) Crystalline $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ or $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$

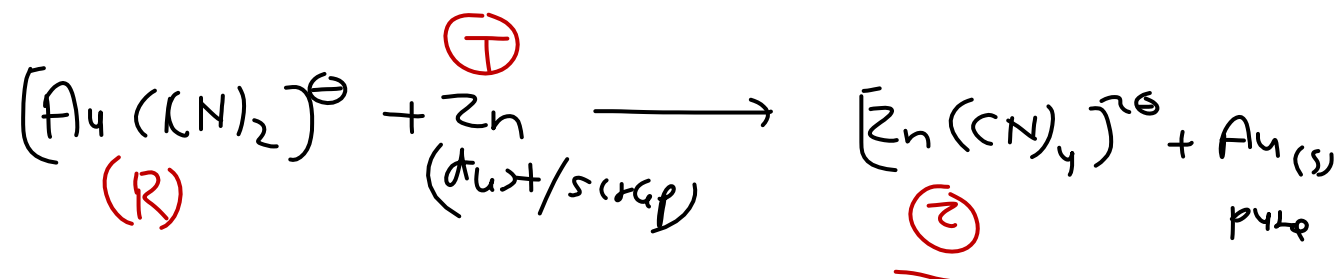
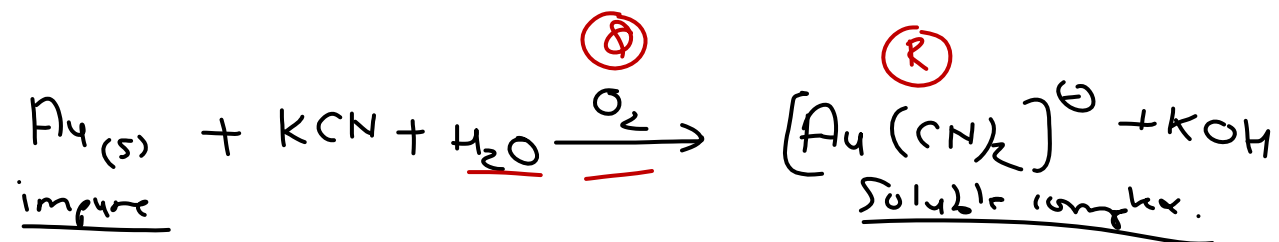


Ans. B, C, D



5. The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in presence of 'Q' in water to form 'R' subsequently, 'R' is treated with 'T' to obtain 'Au' & 'Z'. Choose the correct options.

(A) Z is $[\text{Zn}(\text{CN})_4]^{2-}$ (B) 'T' is 'Zn' (C) R is $[\text{Au}(\text{CN})_2]^-$ (D) 'Q' is O_2



Ans. A, B, C, D

$$[NH_3] = 0.2 M$$

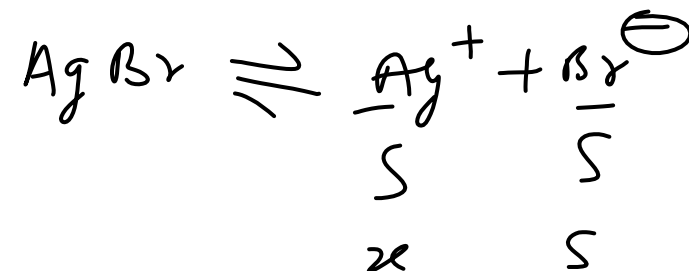
6. NH_3 concentration in a saturated solution of $AgBr(s)$ is maintained at 0.2M. The correct information(s) related with the solution is/are :

$$K_{sp} \text{ of } AgBr = 2 \times 10^{-10}$$

$$K_f \text{ of } Ag(NH_3)_2^+ = 8 \times 10^6$$

$$\lambda_{(Ag^+)}^0 = 6 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

$$\lambda_{(Br^-)}^0 = 8 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

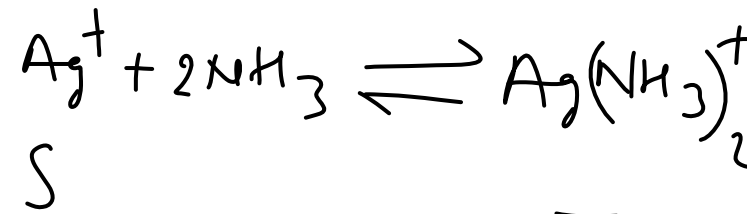


$$(A) \text{ Conductivity of } Br^- = 6.4 \times 10^{-2} \text{ S m}^{-1}$$

$$(B) \text{ Conductivity of } Ag^+ = 1.5 \times 10^{-7} \text{ S m}^{-1}$$

$$(C) [Ag^+] = 8 \times 10^{-3} M$$

$$(D) [Br^-] = 2 \times 10^{-3} M$$



$$\begin{array}{ccc} S & & - \\ 0 & 0.2 M & S \\ x & 0.2+x \approx 0.2 & S-x \approx S \end{array}$$

$$2 \times 10^{-10} = x^2 \quad (1)$$

$$8 \times 10^6 = \frac{S \times x}{x^2 \times 0.2^2}$$

$$8 \times 10^6 = \frac{2 \times 10^{-10}}{x^2 \times 4 \times 10^{-2}}$$

$$x^2 = \frac{1}{16} \times 10^{-14}$$

$$[Ag^+] = x = 2.5 \times 10^{-8} M$$

$$S = \frac{2 \times 10^{-10}}{2.5 \times 10^{-8}} = 8 \times 10^{-3} M$$

$$[Br^-] = S = 8 \times 10^{-3} M$$

$$\lambda^0 = \frac{k}{1000 \times M}$$

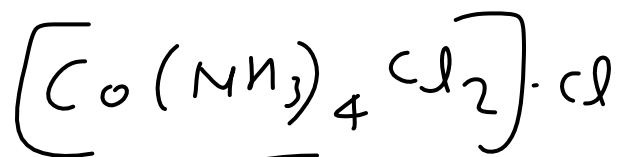
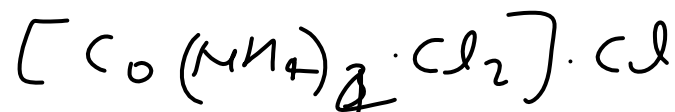
$$k(\text{Ag}^+) = 6 \times 10^{-3} \times 10^{-3} \times 2.5 \times 10^8$$

$$= 1.5 \times 10^{-7} \text{ S cm}^{-1}$$

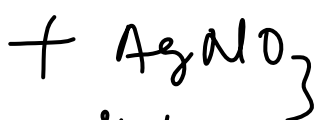
$$k(\text{Br}^-) = 8 \times 10^{-3} \times 10^{-3} \times 8 \times 10^3$$

$$= \underline{6.4 \times 10^{-2} \text{ S cm}^{-1}}$$

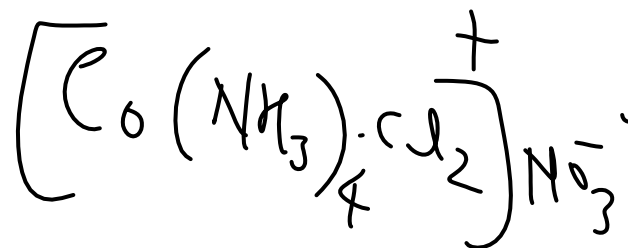
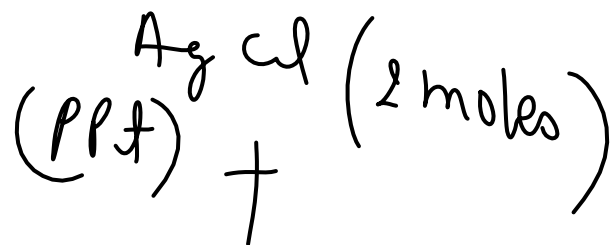
A, B



2 mole



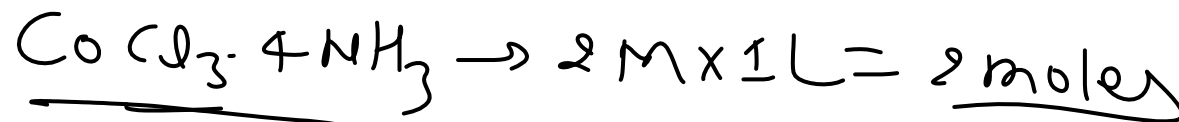
2 mole



2 moles

7. One litre of 2 molar solution of a complex salt $\text{CoCl}_3 \cdot 4\text{NH}_3$ (Mol. wt = 233.5) shows an osmotic pressure of 98.40 atm at 300 K. The solution is now treated with 2 litre of 1M AgNO_3 . Which of the following is/are correct? [Ag = 108] (R = 0.082 L-atm/K-mol)

- ☒ (A) Weight of AgCl precipitated is 287g. $m_{\text{AgCl}} = 2 \times (108 \times 35.5) = 287\text{g}$
☒ (B) The final clear solution will show an osmotic pressure of 98.40 atm.
☒ (C) The final clear solution will show an Osmotic pressure of 32.80 atm.
☒ (D) 2 mole of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \cdot \text{NO}_3$ will be present in solution.



$$\pi = i c R T$$

$$98.4 = i \times 2 \times 0.082 \times 300$$

$$i = \frac{98.4}{49.2} \approx 2$$

$$\pi = i C R T$$

$$= 2 \times \frac{2}{3} \times 0.082 \times 300$$

$$= 4 \times 8.2$$

$$\pi = 32.8 \text{ atm}$$

$$A_1 < D$$

$$k = A e^{-E_a/RT}$$

$$\frac{1.25 \times 10^6}{T^3} = \frac{E_a}{R} \times \frac{1}{T^2} \quad 8.$$

$$\frac{1250 \times 1000}{T} = \frac{E_a}{R}$$

$$E_a(250K) = 2 \times 1250 \times 1000$$

$$E_a = 2500 \text{ Kcal/mol}$$

In order to study effect of temperature on rate of reaction, another term "σ" is defined

as $\sigma = \frac{1}{k} \frac{dk}{dT}$ [where k is rate constant and T represents temperature]. If for a reaction value

of $\sigma = \frac{1.25 \times 10^6}{T^3} \text{ K}^{-1}$ [T should be in Kelvin], then identify the correct statement(s).

- (A) Activation energy for the given reaction at 250 Kelvin is 10 Kcal/mol. ✓
 (B) Activation energy for the given reaction at 2000 Kelvin is 1.25 Kcal/mol. ✓
 (C) Rate of increase of rate constant with temperature is higher at lower temperature range than at higher temperature range. ✓
 (D) The value of $\frac{dk}{dT}$ at 2000 Kelvin is $0.625 \times K$. ✓

$$E_a(2000K) = \frac{2 \times 1250 \times 1000}{2000} = 1.25 \text{ Kcal/mol}$$

$$\frac{dk}{dT} = A \cdot e^{-E_a/RT} \left(+ \frac{E_a}{R} \times \frac{1}{T^2} \right)$$

$$\frac{1}{k} \frac{dk}{dT} = \frac{E_a}{R} \times \frac{1}{T^2}$$

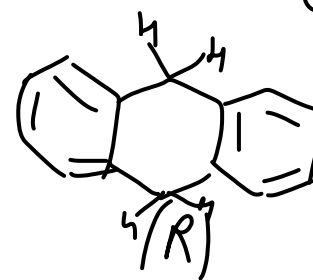
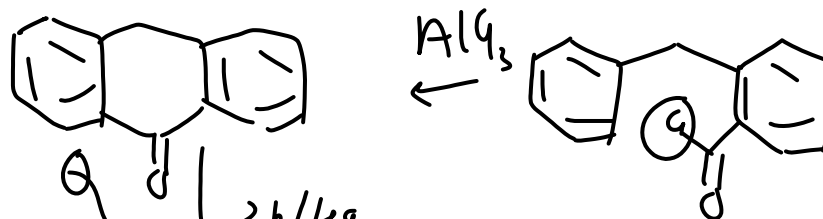
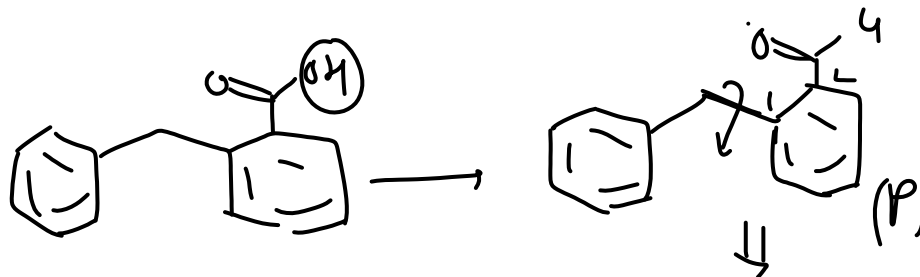
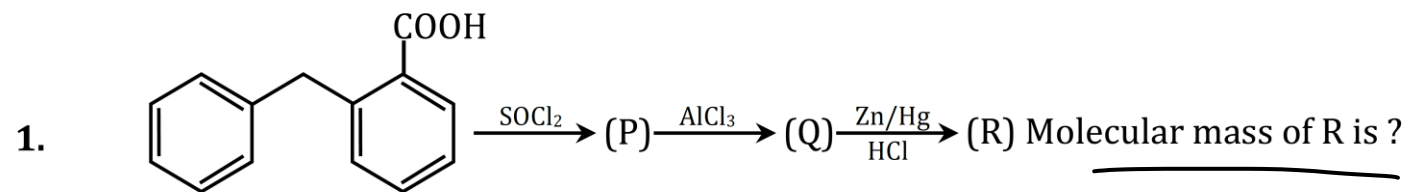
$$\frac{dk}{dT} = ?$$

$$\begin{aligned} \frac{dk}{dT} &= k \sigma = k \times \frac{1.25 \times 10^6}{625 T^3} \\ &= k \times \frac{1250 \times 1000}{2000} \end{aligned}$$

A, B, C

SECTION-III: (Maximum Marks: 18)

- This section contains **SIX (06)** question. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

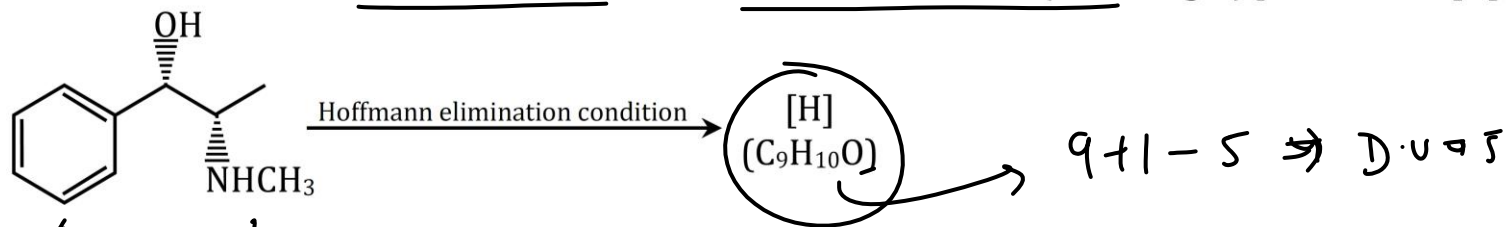


Handwritten calculation for the molecular mass of R:

$$14 \times 12 + 12 = 140 + 12 = 152$$

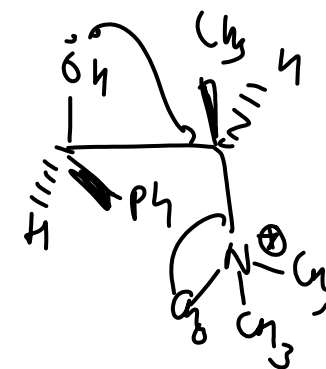
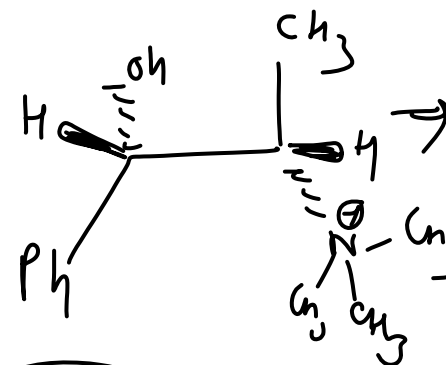
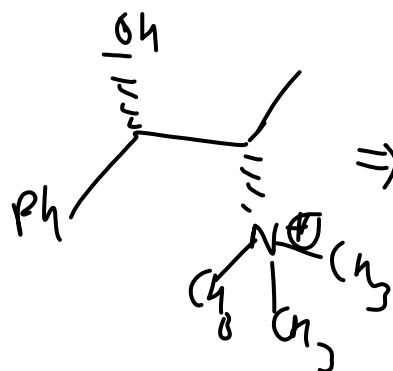
152

2. Pseudoephedrin, a natural alkaloid used as nasal decongestant. When we treat the pseudoephedrin with Hoffman elimination condition gives [H] ($C_9H_{10}O$). However H does not decolorize Br_2 water and does not form an oxime derivative. Number of stereoisomers (including H) possible for [H] is?

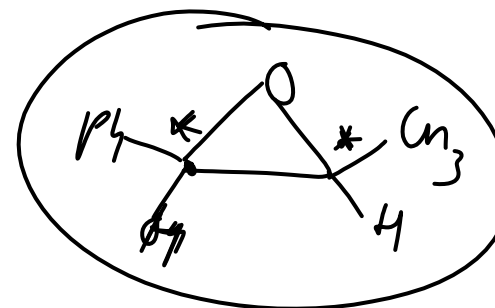


$$9 + 1 - 5 \Rightarrow D.O = 5$$

$$D.O = 4$$



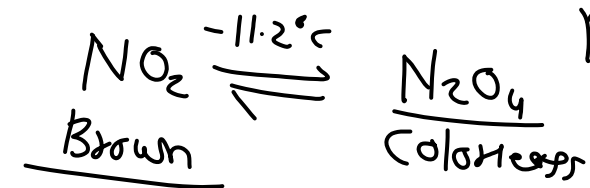
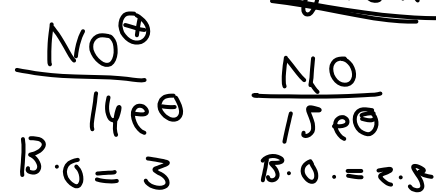
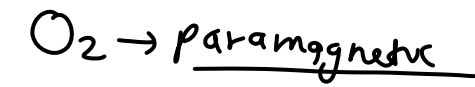
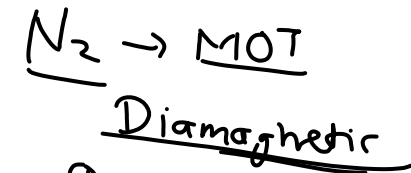
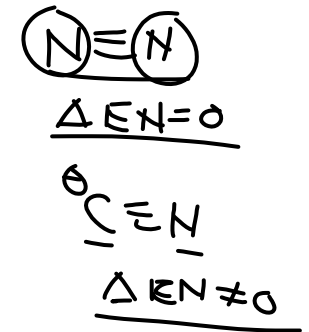
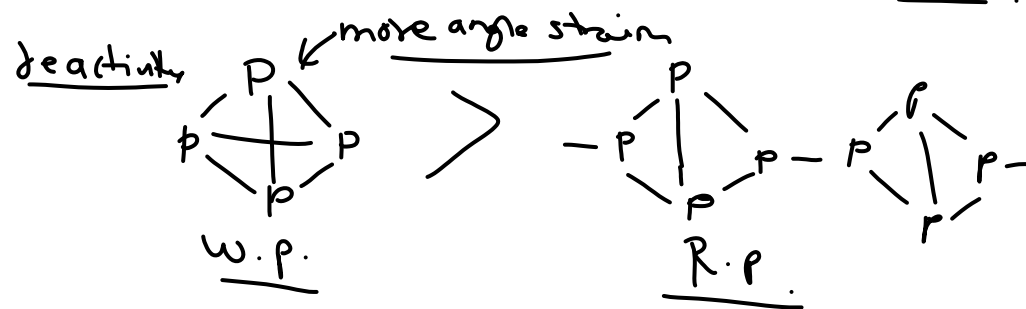
$$h = 2$$



$$2^h = 4$$

3. How many of the following statements are correct reactivity $[H] > H_2$
- ☒ (A) Nascent hydrogen is a powerful reducing agent than molecular hydrogen.
 - ☒ (B) White phosphorous is more reactive than red phosphorous.
 - ☒ (C) The reactivity of K_2O towards water is more than that of Na_2O towards water
 - ☒ (D) Nitrate salt of silver is soluble in ammonia. $[Ag(NH_3)_2]^+$
 - ☒ (E) Liquid nitrogen does not stick to the poles of magnet whereas liquid oxygen does
 - ☒ (F) NO_2 is brown in colour at room temperature but colourless below $0^\circ C$.
 - ☒ (G) Bond order of CN and N_2 are equal but CN is more reactive
 - ☒ (H) NO^+ is having lower bond strength than NO

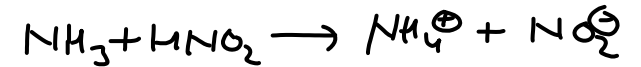
Basic : $K_2O > Na_2O$



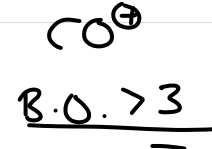
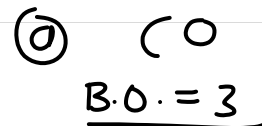
Ans. (5)

4. How many of the following are correct.

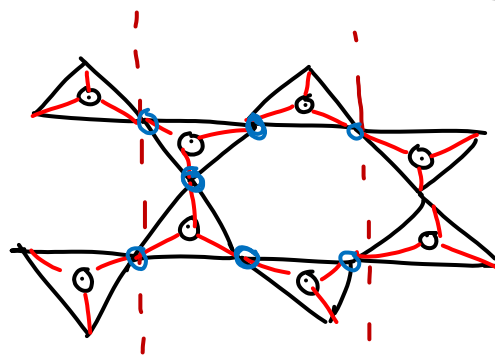
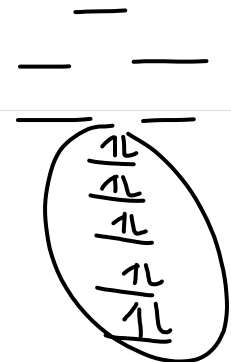
- ✓ (a) Bond lengths order $\text{CO} > \text{CO}^+$ $\text{CO} = 10 \text{ e}^-$
 ✓ (b) If there is no 'sp' mixing, B_2 will be diamagnetic ✓
 ✓ (c) There are 12 five membered rings and 25 six membered rings in C_{70} ✓
 ✓ (d) In double chain silicates some tetrahedra share two corners, some tetrahedra share three corners in such a way that on average 2.5 corners of one tetrahedral are shared.
 ✗ (e) In the reaction of NH_3 with HNO_2 , n-factor values of NH_3 and HNO_2 are not same.
 ✓ (f) dil. HNO_3 on reaction with 'Cu' produces a gas which can form brown ring complex with suitable reagents.
 ✓ (g) order of 'Xe - F' bond lengths $\text{XeF}_2 > \text{XeF}_4 > \text{XeF}_6$



$$n\text{-factor} = 1$$



$\text{B.L.} \propto \frac{1}{\text{B.O.}}$



$\frac{10}{4} = 2.5$
 $[\text{Si}_4\text{O}_{11}]^{6-}$

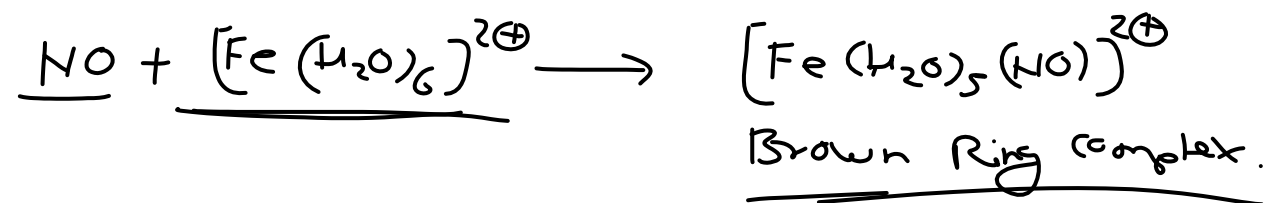
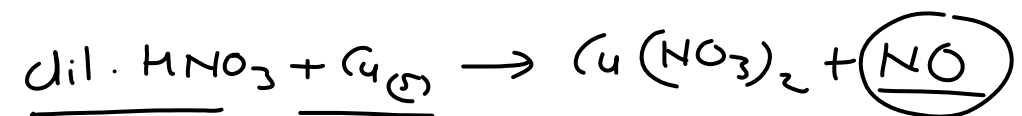
$\text{C}_{70} \Rightarrow 37 \text{ faces}$

25 Hexagonal 12 pentagonal

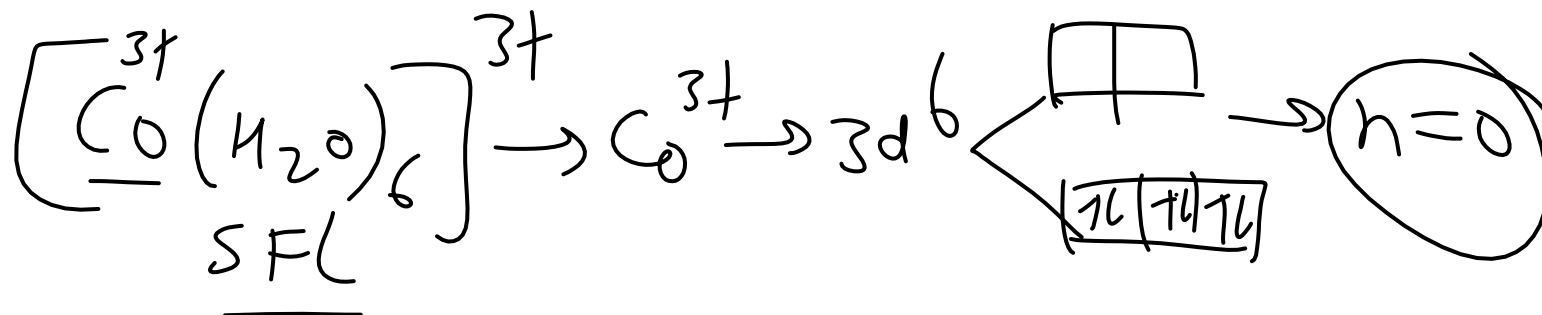
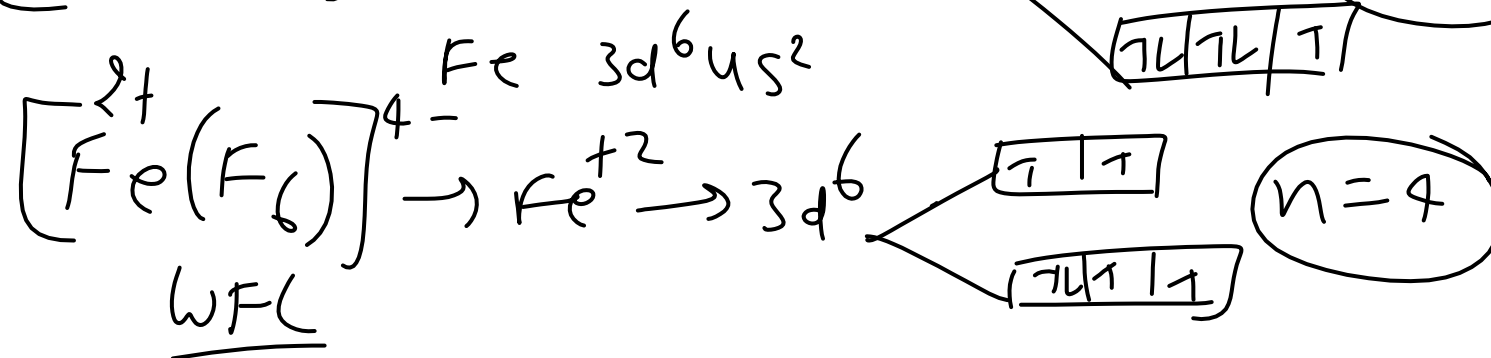
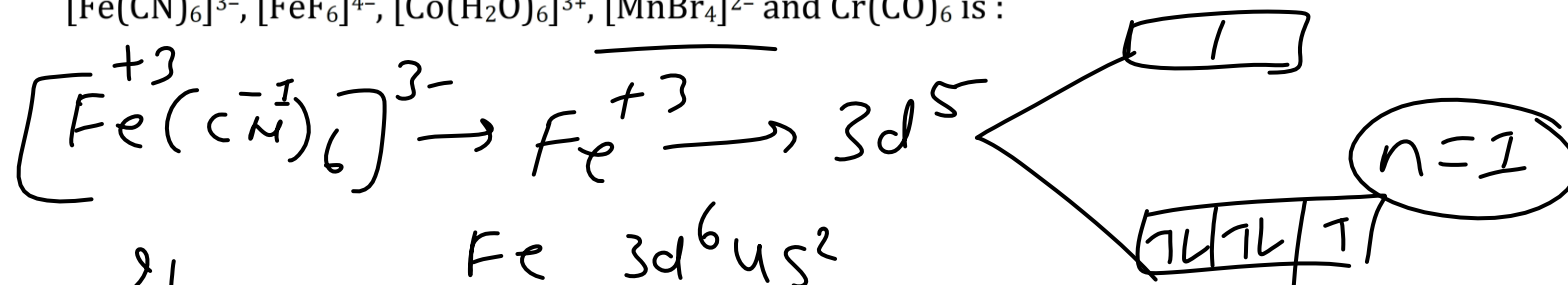
$\text{C}_{60} \Rightarrow 32 \text{ faces}$

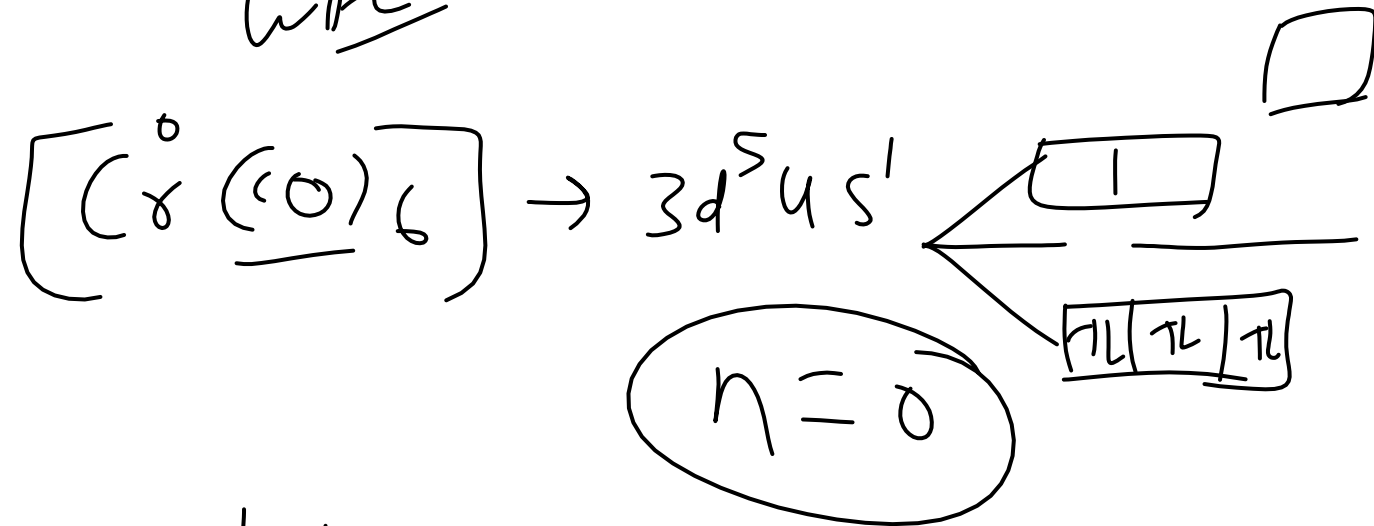
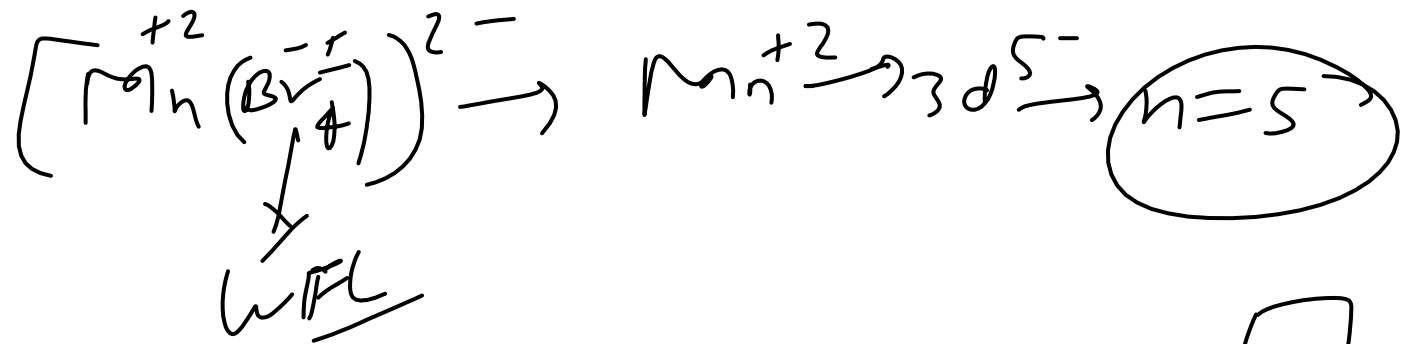
20 hexagonal 12 pentagonal

Ans. (c)



5. The sum of unpaired electrons in the complexes $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{FeF}_6]^{4-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{MnBr}_4]^{2-}$ and $\text{Cr}(\text{CO})_6$ is :





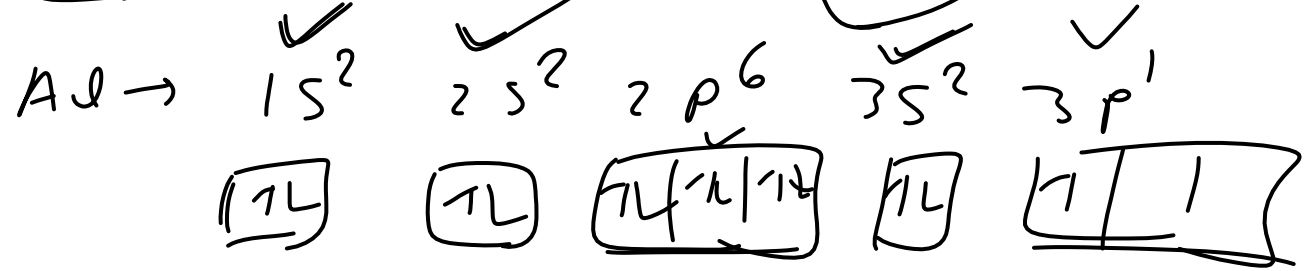
total unpaired = 1 + 4 + 0 + 5 + 0

= 10 e⁻

$$n = 1, 2, 3$$

6.

Find maximum number of electrons in Al for which $\frac{\ell \times m}{n} = 0$. (Atomic number of Al = 13).



ℓ , 0 0 1 0 1

m 0 0 +1 0 -1 0 0 +1 -1

$2 + 2 + 2 + 2 + 1$

$= 9e^-$

JEE

ADVANCED 2024



Mathematics

FINAL PRACTICE
PAPER-1



MATHEMATICS**SECTION-I : (Maximum Marks: 18)**

This section contains **SIX (06)** questions.

Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

For each question, choose the option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

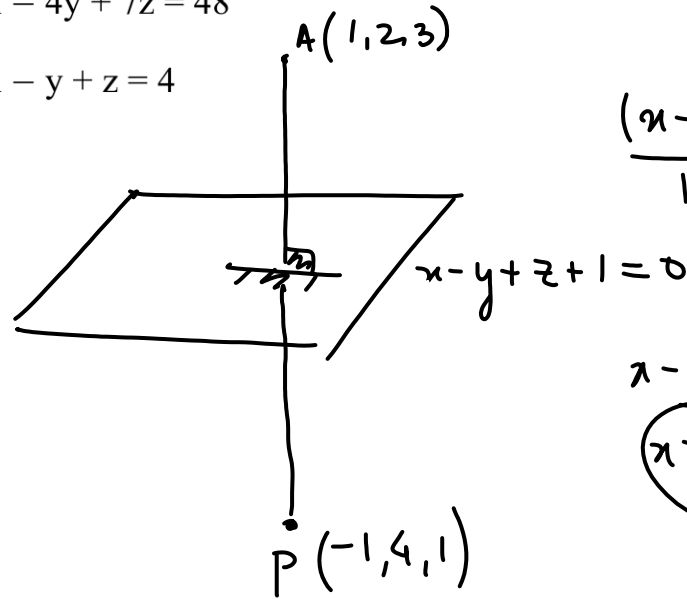
1. Let P be the image of the point $(1, 2, 3)$ with respect to the plane $x - y + z + 1 = 0$. Then the equation of the plane passing through P and containing the line $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-7}{1}$ is $(3, 1, 7) + \lambda(1, 2, 1)$

(A) $x + y - 3z + 17 = 0$

(B) $15x - 2y - 11z + 34 = 0$

(C) $x - 4y + 7z = 48$

(D) $x - y + z = 4$

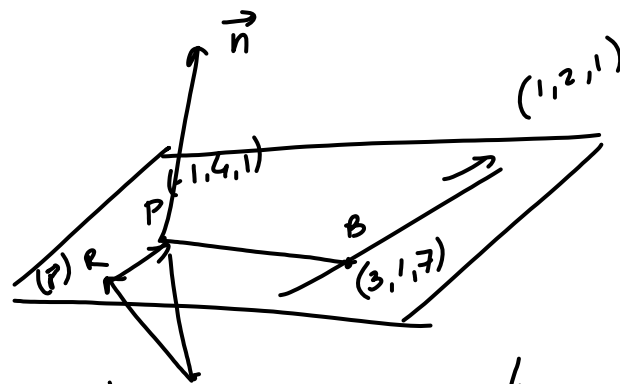


$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \frac{-2(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$$

$$\frac{(x-1)}{1} = \frac{(y-2)}{-1} = \frac{(z-3)}{1} = \frac{-2(\cancel{x}-\cancel{y}+3+1)}{1+1+1} = -2$$

$$x-1 = -2, \quad y-2 = 2, \quad z-3 = -2$$

$$\boxed{x = -1}, \quad \boxed{y = 4}, \quad \boxed{z = 1}$$



$$\vec{PB} = (4, -3, 6) \quad \vec{r} - \vec{p} = (x-1, y-4, z-1)$$

$$(\vec{r} - \vec{p}) \cdot \vec{n} = 0$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 6 \\ 1 & 2 & 1 \end{vmatrix}$$

$$\begin{vmatrix} (x-1) & (y-4) & (z-1) \\ 4 & -3 & 6 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$(x-1)(-3-12) - (y-4)(-2) + (z-1)(8+3) = 0$$

$$-15x - 15 + 2y - 8 + 11z - 11 = 0$$

$$\boxed{15x - 2y - 11z + 34 = 0}$$

2. If $I = \sum_{k=1}^{998} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then

(A) $I < \frac{98}{100} = 0.98$

✓ (C) $I > \frac{99}{100} = 0.99$

(B) $I < \frac{998}{1000} = 0.998$

(D) $I > \ln(999)$

$k < x < k+1$

$$\frac{\cancel{x}}{\cancel{x}(x+1)} < \frac{k+1}{x(x+1)} < \frac{\cancel{k+1}}{x(\cancel{k+1})}$$

$$\int_k^{k+1} \frac{1}{x+1} dx < \int_k^{k+1} \frac{(k+1)}{x(x+1)} dx < \int_k^{k+1} \frac{1}{x} dx$$

$\downarrow$
 I_k

$$\sum_{k=1}^{998} \ln(k+2) - \ln(k+1) < \sum_{k=1}^{998} I_k < \sum_{k=1}^{998} \ln(k+1) - \ln(k)$$

$$I_1 < I < I_2$$

$$I_1 = \begin{array}{l} \cancel{\ln(3)} - \ln(2) \\ \cancel{\ln(4)} - \cancel{\ln(3)} \\ \cancel{\ln(5)} - \cancel{\ln(4)} \\ \vdots \\ \cancel{\ln(999)} - \cancel{\ln(998)} \\ \hline \ln(1000) - \ln(2) \end{array}$$

$$I_2 = \begin{array}{l} \cancel{\ln 2} - \ln 1 \\ \cancel{\ln 3} - \cancel{\ln 2} \\ \vdots \\ \cancel{\ln 999} - \cancel{\ln 998} \\ \hline I_2 = \ln 999 - 0 \end{array}$$

$$I_1 = \ln 1000 - \ln 2$$

$$\begin{array}{ccc}
 \ln 1000 - \ln 2 & < \quad \underline{I} \quad < & \ln(999) \\
 \downarrow & & \downarrow \\
 3 \ln 10 - \ln 2 & & \underline{6.906} \\
 3(2.303) - 0.693 & & \\
 6.909 - 0.693 & & \\
 \underline{6.216} & < \quad \underline{I} \quad < & 6.906
 \end{array}$$

3. If $S = \{1, 2, \dots, 25\}$ and $T = \{x, y\} \subset S$ then find probability that $x^2 - y^2$ is divisible by 7. ($x \neq y$)

(A) $\frac{1}{3}$

(B) $\frac{7}{30}$

(C) $\frac{71}{300}$

~~(D) $\frac{73}{300}$~~

$S = \{1, 2, 3, 4, \dots, 25\}$

$x^2 - y^2 = (x+y)(x-y)$

Total choices = ${}^{25}C_2 = 300$

$(7k_1+1) + (7k_2+1), (7k_1+1) - (7k_2+1)$
 $= 7(k_1+k_2)+2, 7(k_1-k_2)$

$T_1 \rightarrow 7k+1 \Rightarrow 1, 8, 15, 22$
 $T_2 \rightarrow 7k+2 \Rightarrow 2, 9, 16, 23$
 $T_3 \rightarrow 7k+3 \Rightarrow 3, 10, 17, 24$
 $T_4 \rightarrow 7k+4 \Rightarrow 4, 11, 18, 25$
 $T_5 \rightarrow 7k+5 \Rightarrow 5, 12, 19$
 $T_6 \rightarrow 7k+6 \Rightarrow 6, 13, 20$
 $T_7 \rightarrow 7k \Rightarrow 7, 14, 21$

$\Rightarrow 4 \times 4C_2 \{T_1, T_2, T_3, T_4\}$
 $+ 3 \times 3C_2 \{T_5, T_6, T_7\}$
 $+ 2 \times 4C_1 \cdot 3C_1 \{(T_1, T_6), (T_2, T_5)\}$
 $+ 4C_1 \cdot 4C_1 \{T_3, T_4\}$

$$\begin{aligned} \text{favourable cases} &= 4 \times 6 + 3 \times 3 + 2 \times 4 \times 3 + 4 \times 4 \\ &= 24 + 9 + 24 + 16 = 48 + 25 = 73 \end{aligned}$$

$$P(x^2 - y^2 \text{ is divisible by } 7) = \frac{73}{350}$$

$$=$$

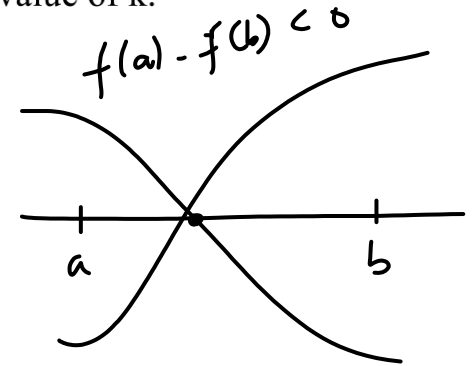
4. Let $\phi(0) = \phi(2a) = \phi(4a) = \phi(6a)$ and $\phi(a) = \phi(3a) = \phi(5a) = \phi(7a)$ but $\phi(0) \neq \phi(a)$. If $\phi(x)$ is twice differentiable function then equation $[\phi'(x+a) - \phi'(x)]^2 + [\phi''(x+a) - \phi''(x)][\phi(x+a) - \phi(x)] = 0$ has atleast 'k' solution in the interval $x \in (0, 6a)$ then find the largest value of k.

(A) 9

(B) 10

(C) 11

(D) 12



$$h(x) = \phi(x+a) - \phi(x)$$

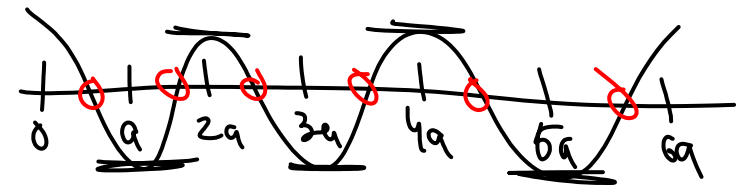
$$h(0) = \phi(a) - \phi(0) = r \quad (r > 0)$$

$$h(a) = \phi(2a) - \phi(a) = \phi(0) - \phi(a) = -r$$

$$h(0) \cdot h(a) < 0 \Rightarrow h(x) = 0 \text{ will have at least one root in } (0, a)$$

$$\underbrace{h(2a) = h(4a) = h(6a) = h(0) = r}_{h(a) \cdot h(2a) < 0}$$

$$h(3a) = h(5a) = h(a) = -r$$



$h(x)=0$ has at least 6 roots in $(0, 6a)$

$h'(x)=0$ " " 5 roots in $(0, 6a)$

$g(x) = h(x) \cdot h'(x) = 0 \longrightarrow$ at least 11 roots in $(0, 6a)$

$g'(x) = (h'(x))^2 + h(x) \cdot h''(x) = 0 \longrightarrow$ 10 roots in $(0, 6a)$

$g'(x) = (\phi'(x+a) - \phi'(x))^2 + (\phi(x+a) - \phi(x))(\phi''(x+a) - \phi''(x)) = 0$

$\rightarrow$ 10 roots.

$k=10$

5. Let P and Q be points of the ellipse $16x^2 + 25y^2 = 400$ so that $PQ = 96/25$ and P and Q lie above major axis. Circle drawn with PQ as diameter touch major axis at positive focus, then the value of slope m of PQ is

(A) -1

(C) 2

Soln

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$(\pm ae, 0)$$

$$(\pm 5(\frac{3}{5}), 0)$$

$$(\pm 3, 0)$$

$$e^2 = 1 - \frac{16}{25}$$

$$e^2 = \frac{9}{25}$$

$$e = \frac{3}{5}$$

$$(3, 0)$$

(B) $\frac{1}{2}$

(D) $\frac{1}{3}$

$$P(5\cos\theta_1, 4\sin\theta_1)$$

$$Q(5\cos\theta_2, 4\sin\theta_2)$$

$$r = \frac{PQ}{2} \Rightarrow \frac{48}{25}$$

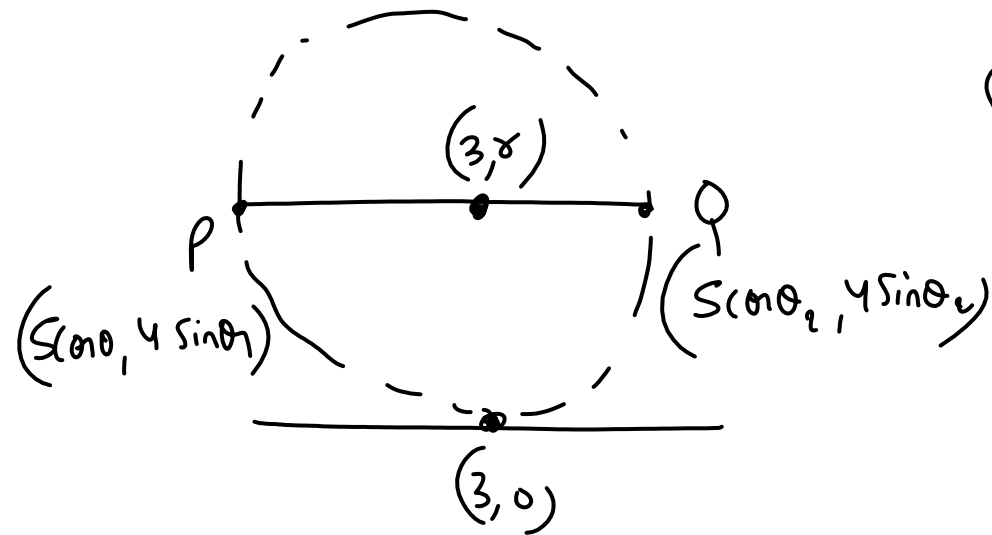
$$m_{PQ} = \frac{4(\sin\theta_2 - \sin\theta_1)}{5(\cos\theta_2 - \cos\theta_1)}$$

$$\Rightarrow \frac{4}{5} \left(\frac{\sin(\theta_1 + \theta_2)}{2} \sin\left(\frac{\theta_2 - \theta_1}{2}\right) \right) / \left(-2\sin\left(\frac{\theta_2 + \theta_1}{2}\right) \sin\left(\frac{\theta_2 - \theta_1}{2}\right) \right)$$

$$m_{PQ} \Rightarrow -\frac{4}{5} \cot\left(\frac{\theta_1 + \theta_2}{2}\right)$$

$$\Rightarrow -\frac{4}{5} \times \frac{12}{5}$$

$$\Rightarrow -\frac{48}{25} \Rightarrow -\frac{48}{25 \times \frac{48}{25}} \Rightarrow -1$$



$$3 = 5 \frac{(\cos \theta_1 + \cos \theta_2)}{2}$$

$$(\cos \theta_1 + \cos \theta_2) = \frac{6}{5}$$

$$\frac{4(\sin \theta_1 + \sin \theta_2)}{2} = 8$$

$$\sin \theta_1 + \sin \theta_2 = \frac{8}{2}$$

$$2 \sin \left(\frac{\theta_1 + \theta_2}{2} \right) \cos \left(\frac{\theta_1 - \theta_2}{2} \right) = \frac{8}{2}$$

$$2 \cos \left(\frac{\theta_1 + \theta_2}{2} \right) \cos \left(\frac{\theta_1 - \theta_2}{2} \right) = \frac{6}{5}$$

$$\cos \left(\frac{\theta_1 + \theta_2}{2} \right) = \frac{\frac{6}{5}}{\frac{4}{2}} \Rightarrow \frac{12}{5} \times \frac{1}{4}$$

6. The area of the region bounded by $\{(x, y); xy \geq -8, -x^2 \leq y \leq -1\}$ is

(A) $8\log_e 2 - \frac{14}{3}$

(B) $16\log_e 2 - \frac{14}{3}$

(C) $16\log_e 2 - 6$

(D) $8\log_e 2 - \frac{7}{3}$

Soln

$xy = -8$

$y = -x^2$
 $x^2 = -y$

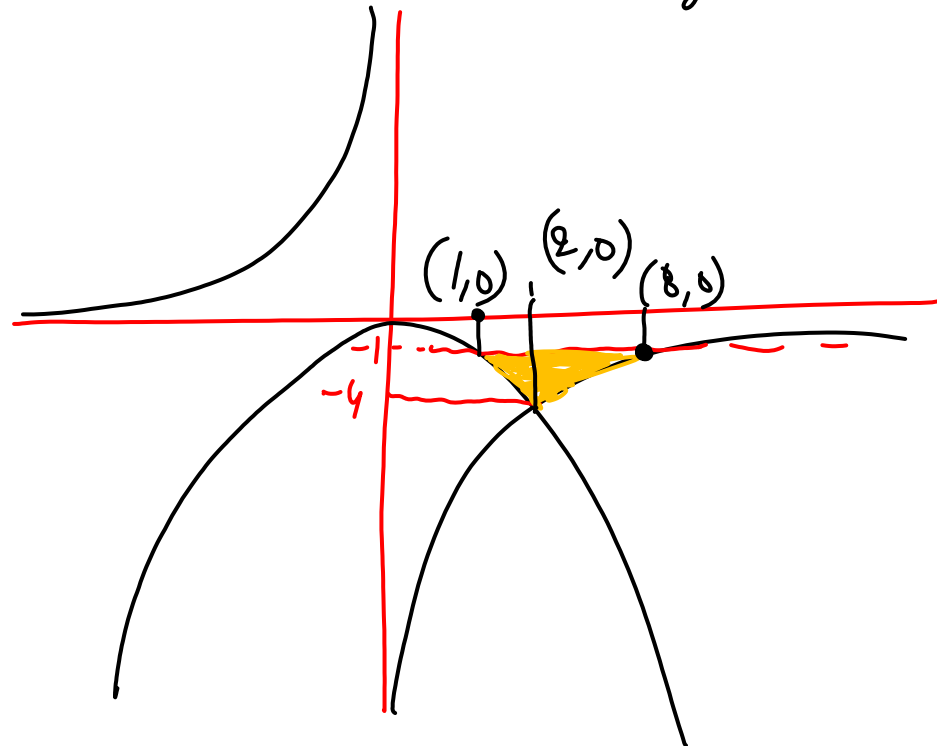
$y = -1$

$xy = -8$
 $y = -x^2$

$-x^3 = -8$

$x = 2$

$y = -4$



$$\int_{-1}^2 (-1 - (-x^2)) dx + \int_2^8 (-1 - (-\frac{8}{x})) dx$$

$$\int_1^2 (x^2 - 1) dx + \int_2^8 \left(\frac{8}{x} - 1 \right) dx$$

$$\left[\frac{x^3}{3} - x \right]_1^2 + \left[8 \ln x - x \right]_2^8$$

$$\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) + \left((8 \ln 8 - 8) - (8 \ln 2 - 2) \right)$$

$$\frac{7}{3} - 1 + (24 \ln 2 - \underline{6} - 8 \ln 2)$$

$$\left(\frac{4}{3} - 6 \right) + 16 \ln 2$$

$$-\frac{14}{3} + 16 \ln 2$$

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider two sets

$$A = \left\{ \frac{p}{q}; p, q \text{ are coprime odd integers} \right\} \text{ and}$$

$$B = \left\{ \frac{p+1}{q}; p, q \text{ are odd integers and } p \neq -1 \right\}$$

Functions $f: A \rightarrow \mathbb{R}$ & $g: B \rightarrow \mathbb{R}$ are both defined as $f(x) = x^x$ & $g(x) = x^{|x|}$ on respective domains.
Then :

$$(A) f\left(\frac{1}{3}\right) \cdot g\left(\frac{-2}{3}\right) = \frac{\sqrt[3]{4}}{3}$$

$$(B) g(x) \text{ is an even function.}$$

$$(C) f(1) + g(2) = 5$$

$$(D) \left[g\left(\frac{-6}{5}\right) \right]^{5/2} - \left[g\left(\frac{4}{5}\right) \right]^{15/4} - \left[f\left(\frac{3}{5}\right) \right]^5 = 1$$

$$g(-x) = g(x)$$

$$x = \frac{p+1}{q} = \frac{\text{even}}{\text{odd}}$$

$$g(x) = (x)^{\frac{\text{even}}{\text{odd}}}$$

$$\Rightarrow \left((x)^{\text{even}} \right)^{\frac{1}{\text{odd}}}$$

$$\Rightarrow g(-x) = g(x)$$

↓
even

$$(A) f\left(\frac{1}{3}\right) \cdot g\left(\frac{-2}{3}\right)$$

$$\Rightarrow \left(\frac{1}{3}\right)^{\frac{1}{3}} \left(\frac{-2}{3}\right)^{\frac{1}{3}} \Rightarrow \left(\frac{1}{3}\right)^{\frac{1}{3}} \left(\frac{4}{9}\right)^{\frac{1}{3}} \Rightarrow \left(\frac{4}{27}\right)^{\frac{1}{3}} \Rightarrow \frac{4}{3}$$

$$\left[9 \left(-\frac{6}{5} \right) \right]^{\frac{5}{2}} - \left[9 \left(\frac{4}{5} \right) \right]^{\frac{15}{4}} - \left[6 \left(\frac{3}{5} \right) \right]^5 \quad g(x) = x^{1/x}$$

$$\left(\left(-\frac{6}{5} \right)^{\frac{6}{5}} \right)^{\frac{5}{2}} - \left(\left(\frac{4}{5} \right)^{\frac{4}{5}} \right)^{\frac{15}{4}} - \left(\left(\frac{3}{5} \right)^{\frac{3}{5}} \right)^5 \quad f(x) = x^x$$

$$\left(\left(-\frac{6}{5} \right)^{\frac{6}{5}} \right)^{\frac{5}{2}} -$$

$$\left(\left(-\frac{6}{5} \right)^6 \right)^{\frac{1}{2}}$$

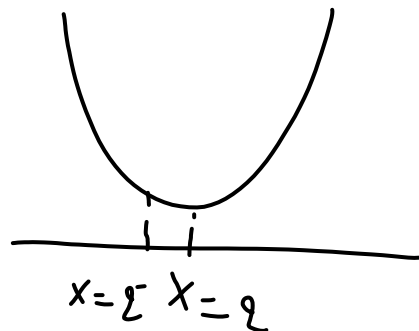
$$\left(\frac{6}{5} \right)^3 - \left(\frac{4}{5} \right)^3 - \left(\frac{3}{5} \right)^3 \Rightarrow \frac{216 - 64 - 27}{125} \Rightarrow \frac{125}{125} \Rightarrow (1)$$

2. Let the function $f(x) = \begin{cases} 2x - x^3 + \ln(a^2 - 3a + 3), & 0 \leq x < 2 \\ x - 6, & x \geq 2 \end{cases}$. It possess a local minimum at $x = 2$

for

(A) ~~$a = 0$~~

(C) ~~$a = 4$~~



(B) ~~$a = 3$~~

(D) ~~$a = -1$~~

$$f(2^-) \geq f(2)$$

$$a^2 - 3a + 2 \geq 0$$

$$(a-1)(a-2) \geq 0$$

+	-	+
	1	2

$$f(2) - (2)^3 + \ln(a^2 - 3a + 3) \geq (2-6) \quad a \in (-\infty, 1] \cup [2, \infty)$$

$$-4 + \ln(a^2 - 3a + 3) \geq -4$$

$$\ln(a^2 - 3a + 3) \geq 0$$

$$a^2 - 3a + 3 \geq 1$$

3. If $(1 + x + x^2)^{100} = a_0 + a_1x + a_2x^2 + \dots + a_{200}x^{200}$, then the INCORRECT relation from among the following is/are:

(A) $37 a_{37} = 64 a_{36} + 165 a_{35}$

(B) $36 a_{37} = 165 a_{35} + 64 a_{36}$

(C) $38 a_{37} = 36 a_{35} + 37 a_{36}$

(D) $74 a_{37} = 37 a_{36} + 64 a_{35}$

$$(1+x+x^2)^{100} = a_0 + a_1x + a_2x^2 + \dots + a_{35}x^{35} + a_{36}x^{36} + a_{37}x^{37} + \dots + a_{200}x^{200}$$

diff wrt x

$$100(1+x+x^2)^{99}(1+2x) = a_1 + 2a_2x + \dots + 35a_{35}x^{34} + 36a_{36}x^{35} + 37a_{37}x^{36} + \dots + 200a_{200}x^{199}$$

multiply with $(1+x+x^2)$

$$100(1+2x)(1+x+x^2)^{100} = (1+x+x^2)(a_1 + 2a_2x + \dots + 35a_{35}x^{34} + 36a_{36}x^{35} + 37a_{37}x^{36} + \dots + 200a_{200}x^{199})$$

$$100(1+2x)(a_0 + a_1x + a_2x^2 + \dots + a_{35}x^{35} + a_{36}x^{36} + a_{37}x^{37}) = (1+x+x^2)(a_1 + 2a_2x + \dots + 35a_{35}x^{34} + 36a_{36}x^{35} + 37a_{37}x^{36} + \dots)$$

coeff of x^{36}

$$1a(a_{36} + 2a_{35}) = 37a_{37} + 36a_{36} + 35a_{35}$$

$$1a a_{36} + 2a a_{35} = 37a_{37} + 36a_{36} + 35a_{35}$$

$$37a_{37} = 64a_{36} + 165a_{35}$$

Correct

4. Let $f(x)$ is twice differentiable and increasing function such that domain of $g(x) = \sqrt{f(x) - x}$ is $\mathbb{R}$, $h(x) = \frac{1}{1-x}$ and $\phi(x) = \sqrt{f(f(f(x))) - h(h(h(x)))}$. If α, β are two distinct roots of $g(x) = 0$ and both are greater than 1, then which of the following statement is/are CORRECT?

~~(A)~~ domain of $\phi(x)$ is $\mathbb{R} - \{0, 1\}$

~~(B)~~ Range of $\phi(x)$ is always $\mathbb{R}$

~~(C)~~ If α is repeated root of $g(x) = 0$, then it is also repeated root of $\phi(x) = 0$

~~(D)~~ If $f(f(f(x)))$ is twice differentiable function, then $\phi''(x) = 0$ always has two distinct roots

$f(x) \uparrow$

$$g(x) = \sqrt{f(x) - x} \quad x \in \mathbb{R} \quad h(x) = \frac{1}{1-x} \quad x \neq 1$$

$\alpha \quad \beta \quad (\alpha, \beta > 1)$

$$h(h(x)) = \frac{1}{1 - \frac{1}{1-x}}, x \neq 1$$

$$h(h(x)) = \frac{1-x}{-x}$$

$$h(h(x)) = \frac{x-1}{x}, x \neq 1, 0$$

$$h(h(h(x))) = \frac{\frac{1}{1-x} - 1}{1 - \frac{1}{1-x}} = \frac{\frac{x}{1-x}}{\frac{-x}{1-x}} = x$$

$$\phi(x) = \sqrt{F(F(f(x))) - x}$$

$$g(x) \geq 0$$

$$f(x) - x \geq 0$$

$$f(x) \geq x$$

$$\Rightarrow f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) \geq f(f(x)) \geq f(x)$$

$$\Rightarrow f(f(f(x))) \geq f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) \geq x$$

$$\underline{f(f(f(x))) - x \geq 0}$$

$$\underline{f(f(f(x))) - x \geq 0}$$

Domain of $g(x)$

$$\underline{x \in \mathbb{R} - \{9, 1\}}$$

$$f(x) = x$$

$$f(f(x)) = f(x)$$

$$f(f(x)) = x$$

$$f(f(f(x))) = f(x)$$

$$f(f(f(x))) = x$$

$$\underline{f(f(f(x))) - x = 0}$$

$$\underline{\phi(x) = 0}$$

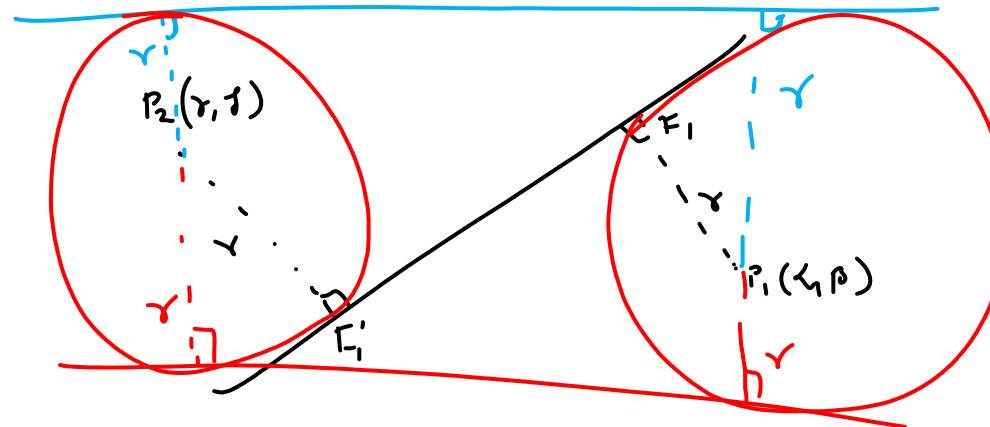
5. Let $P_1(\alpha, \beta)$ and $P_2(\gamma, \delta)$ are the only two points which is at a distance of ' r ' ($r \neq 0$) units from the lines $L_i : a_i x + b_i y + c_i = 0$ ($i = 1, 2, 3$) where $(a_1 \alpha + b_1 \beta + c_1)(a_1 \gamma + b_1 \delta + c_1) < 0$. Foot of perpendicular from P_1 and P_2 on L_i is F_i, F_i' respectively ($i = 1, 2, 3$)

~~(A)~~ $F_2 F_2' = \sqrt{(\alpha - \gamma)^2 + (\beta - \delta)^2}$

~~(B)~~ $F_1 F_1' = 2r \left| \frac{a_1 b_2 - a_2 b_1}{a_1 a_2 + b_1 b_2} \right|$

(C) $(F_2 F_2')^2 - (F_1 F_1')^2 = 2r^2$

~~(D)~~ $(F_2 F_2')^2 - (F_1 F_1')^2 = 4r^2$



$P_1(\alpha, \beta)$
 $P_2(\gamma, \delta)$
 $a_1 x + b_1 y + c_1 = 0 \rightarrow L_1$
 $a_2 x + b_2 y + c_2 = 0 \rightarrow L_2$
 $a_3 x + b_3 y + c_3 = 0 \rightarrow L_3$

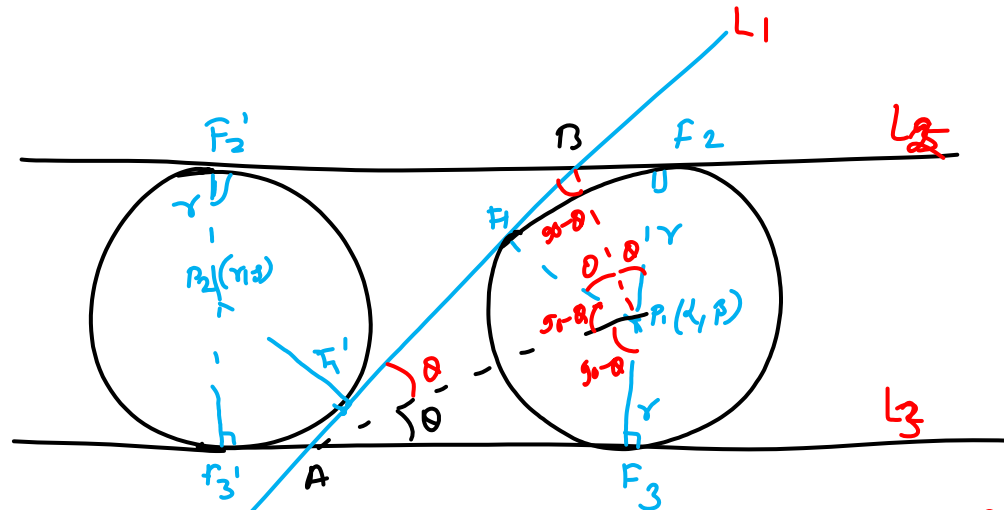
$-4r^2 \cos^2 2\theta + 4r^2 \cos^2 2\theta$
 $= 4r^2$

2 α angle b/w
line L_1 & L_3

$$\tan 2\theta = \left| \frac{-\frac{a_3}{b_3} + \frac{a_1}{b_1}}{1 + \frac{a_3 a_1}{b_3 b_1}} \right|$$

$$= \left| \frac{a_1 b_3 - a_3 b_1}{b_3 b_1 + a_3 a_1} \right|$$

$$= \frac{a_1 b_2 - a_2 b_1}{b_2 b_1 + a_2 a_1}$$



$$\frac{r}{AF_3} = \tan \theta$$

$$AF_3 = r \cot \theta = AF_1$$

$$\frac{r}{BF_1} = \tan(\theta - \theta) \quad BF_1 = r \tan \theta = BF_2 = AF_1'$$

$$F_1 F_1' = r \cot \theta - r \tan \theta$$

$$= r \frac{\cos \theta}{\sin \theta} - \frac{r \sin \theta}{\cos \theta}$$

$$= \frac{2r \cos 2\theta}{\sin 2\theta} = 2r \cot 2\theta$$

$$F_2 F_2' = BF_2' + BF_2$$

$$= BF_1 + F_1 F_1' + r \tan \theta$$

$$= r \tan \theta + r \cot \theta - r \tan \theta + r \tan \theta$$

$$= r \cot \theta + r \tan \theta$$

$$= 2r \csc 2\theta$$

$$\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} = \frac{1}{\sin \theta \cos \theta}$$

$$F_2 F_2' = \sqrt{(x-r)^2 + (y-r)^2}$$

6. Five fair dice are tossed. Which of the following is/are true ?

~~(A)~~ mean of number of dice showing even numbers is 2.5

~~(B)~~ variance of number of dices showing even numbers is 1.25

~~(C)~~ probability that same numbers appeared on all the 5 dice is $\frac{1}{6^4}$

~~(D)~~ probability that atleast two dice show even number, if atleast one die show even number is $\frac{26}{31}$

Solⁿ $x = \text{number of dice showing even numbers}$

$x = 0$	${}^5C_0 \cdot \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$
$x = 1$	${}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{5}{32}$
$x = 2$	${}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32}$
$x = 3$	${}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32}$
$x = 4$	${}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{32}$
$x = 5$	${}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32}$

$$P(e) = \frac{1}{2} = P$$

$$P(u) = \frac{1}{2} = P$$

$$\begin{aligned}\text{mean} &= np \\ &= 5 \times \frac{1}{2} = 2.5\end{aligned}$$

$$\begin{aligned}\text{variance} &= npq \\ &= 5 \times \frac{1}{2} \times \frac{1}{2} = 1.25\end{aligned}$$

$$\begin{aligned}\textcircled{c} \text{ same number on all the dice} &= \frac{6}{6^5} \\ &= \frac{1}{6^4}\end{aligned}$$

$$\textcircled{d} \quad \frac{1 - \frac{1}{32} - \frac{5}{32}}{1 - \frac{1}{32}} \Rightarrow \frac{\frac{26}{32}}{\frac{31}{32}} = \frac{26}{31}$$

7. Consider a pyramid OPQRS lying in 1st octant with base OPQR where O, P, Q, R are (0,0,0), (4,0,0), (6,0,4) & (0,0,4) respectively. 'S' is a point 4 units above the base such that it's exactly above M (i.e. SM $\perp$ base) where OM + PM + QM + RM is minimum. Then

① M pt coordinate

OQ & PR

OQ $\Rightarrow$

$$\frac{x-0}{6} = \frac{y-0}{0} = \frac{z-0}{4} = 1$$

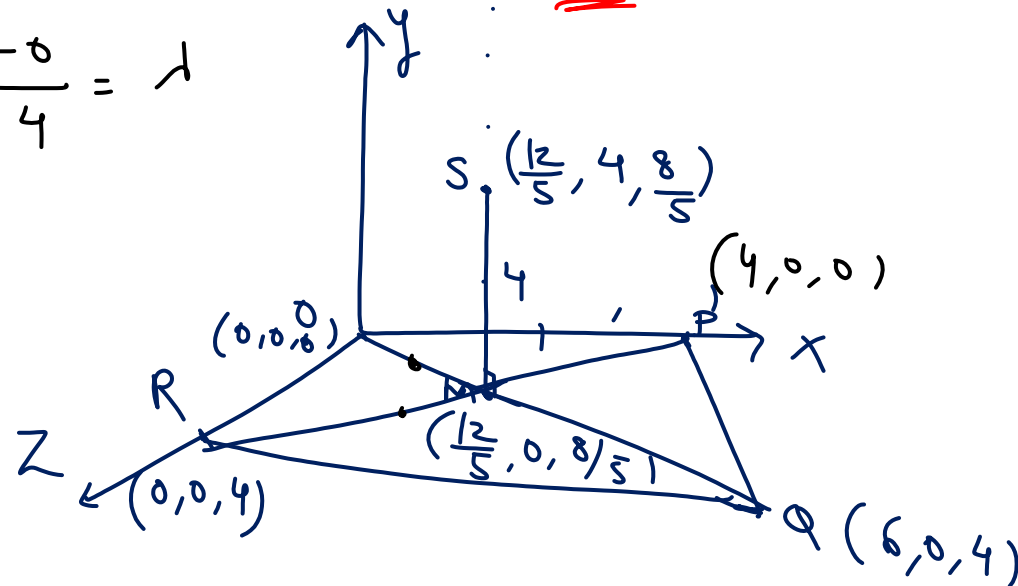
$$(6, 0, 4)$$

(A) Plane containing O, R & S is $x - \frac{3y}{5} = 0$

(C) Perpendicular distance from Q to the plane containing P, R, S is $5\sqrt{2}$ units

(B) Plane containing P, R & S is $x + z = 4$

(D) Acute angle between OS & OQ is $\frac{\pi}{3}$



M pt should
pt of intersection
of diag.

Eqⁿ of PR : $R(0, 0, 4), P(4, 0, 0)$

$$\frac{x-0}{4} = \frac{y-0}{0} = \frac{z-4}{-4} = \mu$$

$$(4\mu, 0, -4\mu+4) \equiv (6\lambda, 0, 4\lambda)$$

$$6\lambda = 4\mu \quad \& \quad 4\lambda = -4\mu + 4$$

$$\boxed{3\lambda = 2\mu}$$

$$2\lambda = -2\mu + 2$$

$$2\lambda = -3\lambda + 2$$

$$\boxed{\lambda = 2/5}$$

$$M\left(\frac{12}{5}, 0, \frac{8}{5}\right)$$

Eqⁿ $P, R, S \rightarrow$

$\downarrow$

$$(4, 0, 0), R(0, 0, 4)$$

$$S\left(\frac{12}{5}, 4, \frac{8}{5}\right)$$

Eqⁿ : $\boxed{x+z=4} \rightarrow \textcircled{1}$

$$\begin{aligned} & \text{Distance from } (4, 0, 0) \text{ to } S\left(\frac{12}{5}, 4, \frac{8}{5}\right) \\ &= \sqrt{\left(4 - \frac{12}{5}\right)^2 + (0-4)^2 + \left(0 - \frac{8}{5}\right)^2} \\ &= \sqrt{\left(\frac{8}{5}\right)^2 + 16 + \left(\frac{64}{25}\right)} \\ &= \sqrt{\frac{64}{25} + \frac{400}{25} + \frac{64}{25}} \\ &= \sqrt{\frac{464}{25}} = \frac{\sqrt{464}}{5} = \frac{4\sqrt{29}}{5} \end{aligned}$$

$$O(0,0,0)$$

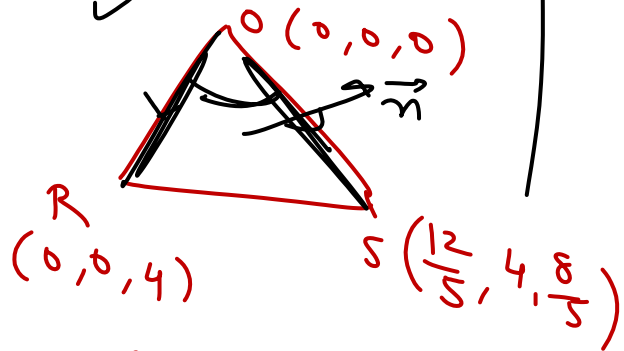
$$P(4,0,0)$$

$$Q(6,0,4)$$

$$R(0,0,4)$$

$$S\left(\frac{12}{5}, 4, \frac{8}{5}\right)$$

(i) Eq Plane... ORS



$$a(x-0) + b(y-0) + c(z-0) = 0$$

$$ax + by + cz = 0$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4 \\ \frac{12}{5} & 4 & \frac{8}{5} \end{vmatrix}$$

$$\vec{n} = \hat{i}(-16) - \hat{j}\left(-\frac{48}{5}\right) + \hat{k}(0)$$

$$\text{Eqn} \therefore -16x + \frac{48}{5}y + 0 \cdot z = 0$$

$$x - \frac{3}{5}y = 0$$

$$OS \triangle OQ$$

$$\underline{\underline{dr's}} \quad \frac{12}{5}, 4, \frac{8}{5}$$

$$\vec{OS} \text{ dr's} \therefore (6, 0, 4)$$

$$\cos \theta = \frac{\frac{12}{5} \times 6 + 0 + \frac{32}{5}}{5}$$

$$\sqrt{\frac{144}{25} + 16 + \frac{64}{25}} \sqrt{36 + 0 + 16}$$

$$\cos \theta = \sqrt{\frac{13}{38}} \neq \frac{1}{2}$$

$$\theta \neq \frac{\pi}{3}$$

$$A \rightarrow \vec{OA} = \vec{a} + 2\vec{b}$$

$$B \rightarrow \vec{OB} = \vec{a} + 3\vec{b}$$

$$C \rightarrow \vec{OC} = \vec{a} + 5\vec{b}$$

$$\vec{AB} = \vec{BC}$$

$$\vec{AB} = \vec{b}$$

$$\vec{BC} = 2\vec{b}$$

$$\Rightarrow \vec{AB} = \frac{1}{2} \vec{BC}$$

8. Let L is a line and A is foot of perpendicular from O (origin) on L. P and Q are two other points on line L. Position vectors of A, P, Q are $\vec{a}, \vec{x}, \vec{y}$ respectively. If $|\vec{a}| = 1$, centroid of $\triangle OPQ$ is $\vec{c}$ and $\vec{b} = \vec{a} \times (\vec{x} \times \vec{y})$, then

(A) $\vec{x} = \frac{\vec{b} + 3\vec{c}}{2}$

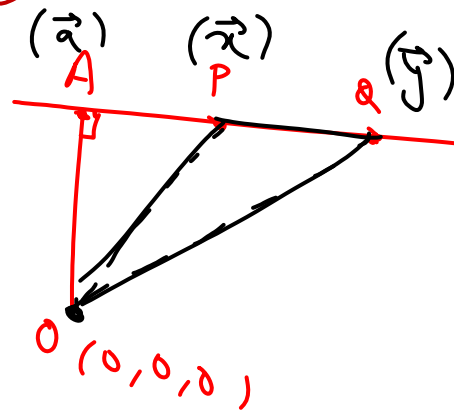
(B) $\vec{y} = \frac{3\vec{c} - \vec{b}}{2}$

(C) $\vec{b}$ is parallel to line L

(D) $\vec{a} + 2\vec{b}, \vec{a} + 3\vec{b}, \vec{a} + 5\vec{b}$ are position vectors of collinear points.

$$|\vec{a}| = 1$$

$$\vec{c} = \frac{\vec{x} + \vec{y} + \vec{o}}{3}$$



$$\Rightarrow 3\vec{c} = \vec{x} + \vec{y}$$

$$\vec{b} = \vec{a} \times (\vec{x} \times \vec{y}) = (\vec{a} \cdot \vec{y}) \vec{x} - (\vec{a} \cdot \vec{x}) \vec{y}$$

$$\checkmark \boxed{\vec{b} = \vec{x} - \vec{y}} \text{ --- (1)} \quad \boxed{3\vec{c} = \vec{x} + \vec{y}} \text{ --- (2)}$$

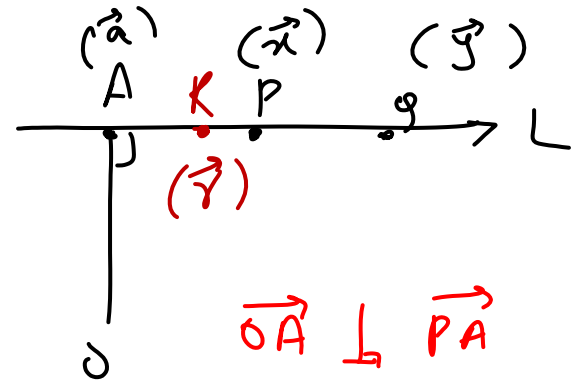
$$2\vec{x} = \vec{b} + 3\vec{c} \quad \vec{y} = \frac{3\vec{c} - \vec{b}}{2}$$

$$\vec{x} = \frac{\vec{b} + 3\vec{c}}{2}$$

Eqⁿ of line: $(\vec{r} - \vec{a}) \parallel (\vec{y} - \vec{x})$

$$\vec{r} = \vec{a} + k(\vec{x} - \vec{y})$$

$$\vec{r} = \vec{a} + k\vec{b}$$



$$\vec{OA} \perp \vec{PA}$$

$$\vec{a} \cdot (\vec{a} - \vec{x}) = 0$$

$$|\vec{a}|^2 - \vec{a} \cdot \vec{x} = 0$$

$$\vec{a} \cdot \vec{x} = 1$$

$$\vec{OA} \perp \vec{AB}$$

$$\vec{a} \cdot \vec{y} = 1$$

SECTION-III : (Maximum Marks: 18)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33 , $-.30$, 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $ax^4 + bx^3 + x^2 + (3-a)x + 3 = 0$ and $x^2 + (2-a)x + 3 = 0$ have common roots. If $a \in (-1, 5)$ then find

$|a - 12b|$

$= 2.50$

$\rightarrow$ nature

$D = B^2 - 4AC$

$= (2-a)^2 - 4(1)(3)$

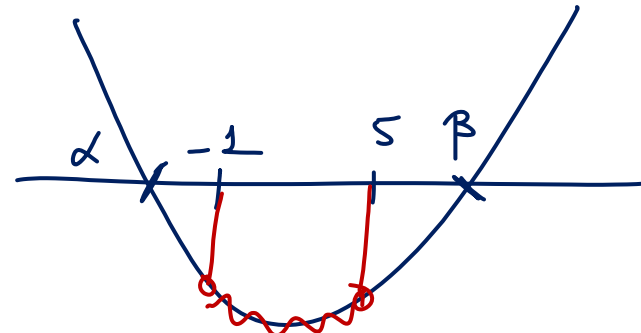
$D = a^2 - 4a - 8$

$\alpha = 2 - 2\sqrt{3}$

$\beta = 2 + 2\sqrt{3}$

$a = \frac{4 \pm \sqrt{16 + 32}}{2(1)}$

$a = 2 \pm 2\sqrt{3}$



$D < 0$

Roots are neg $\Rightarrow$ Pair $\rightarrow$ Conjugate

Both roots of $x^2 + (2-a)x + 3 = 0$ are common.

$$ax^4 + bx^3 + x^2 + (3-a)x + 3 = (x^2 + (2-a)x + 3)(ax^2 + cx + 1)$$

① coeff of x :- $3-a = 2-a + 3c \Rightarrow c = \frac{1}{3}$

② coeff of x^2 :- $1 = 1 + 3a + (2-a)\frac{1}{3}$
 $3a + \frac{2}{3} - \frac{a}{3} = 0 \Rightarrow \frac{8a}{3} = -\frac{2}{3} \Rightarrow a = -\frac{1}{4}$

③ coeff of x^3 :-

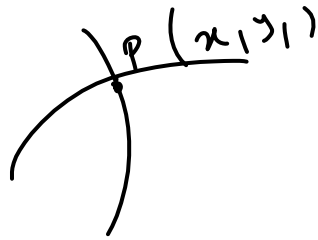
$$b = c + a(2-a)$$

$$b = \frac{1}{3} - \frac{1}{4}\left(2 + \frac{1}{4}\right)$$

$$= \frac{1}{3} - \frac{9}{16} = \frac{16-27}{48}$$

$$b = -\frac{11}{48}$$

$$\begin{aligned} & |a - 12b| \\ & \left| -\frac{1}{4} + 12 \times \frac{11}{48} \right| \\ & = \left| \frac{10}{4} \right| = \frac{5}{2} \\ & = 2.50 \end{aligned}$$



2. Let the line $y = mx$ and $\log(x^2 + y^2) = 2 \left(\cot \sqrt{\frac{\pi^2}{16} - \alpha^2} \right) \tan^{-1} \left(\frac{y}{x} \right)$ intersect each other at an acute angle θ . If maximum value of θ is equal to $b\pi$ then b is equal to

angle B/w Two curves -

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

(1) $y = mx$ → Slope - m

$P(x_1, y_1) \Rightarrow m = \frac{y_1}{x_1} \rightarrow m_2$

$$\frac{1}{x^2 + y^2} (x + yy') = K \left\{ \frac{1}{1 + y^2/x^2} \right\} \left\{ \frac{xy' - y(1)}{x^2} \right\}$$

$$\Rightarrow \frac{x + yy'}{x^2 + y^2} = K \left\{ \frac{x^2}{x^2 + y^2} \right\} \left\{ \frac{xy' - y}{x^2} \right\}$$

$$\frac{dy}{dx} \{ y - Kx \} = -Ky - x$$

$$\frac{dy}{dx} = \frac{x+ky}{kx-y} \bigg|_{P(x_1, y_1)}$$

$$= \frac{x_1 + ky_1}{kx_1 - y_1} \rightarrow m_1$$

$$\tan \theta = \left| \frac{\frac{x_1 + ky_1}{kx_1 - y_1} - \frac{y_1}{x_1}}{1 + \left(\frac{x_1 + ky_1}{kx_1 - y_1} \right) \left(\frac{y_1}{x_1} \right)} \right|$$

$$\rightarrow \tan \theta = \left| \frac{x_1^2 + kx_1y_1 - kx_1y_1 + y_1^2}{kx_1^2 - x_1y_1 + x_1y_1 + ky_1^2} \right|$$

$$\tan \theta = \left| \frac{x_1^2 + y_1^2}{k(x_1^2 + y_1^2)} \right| = \left| \frac{1}{k} \right|$$

$$\theta_{\max} = \frac{\pi}{4} = 67^\circ$$

$$b = \frac{1}{4}$$

$$b = 0.25$$

$$\tan \theta = \tan \sqrt{\frac{\pi^2}{16} - \alpha^2}$$

$$\theta = \sqrt{\frac{\pi^2}{16} - \alpha^2}$$

Max $\alpha = 0$

3. Let all the sub matrices of order 2×2 of matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ have their determinant equals to zero

where $a, b, c, d, e, f, g, h, i \in \{1, 2, 3, 4, 5, 6\}$. If no two elements of same row are same then the total number possible matrixes of A. is/are 156

Soln $\rightarrow$ $\begin{vmatrix} a & b \\ d & e \end{vmatrix} = 0, \begin{vmatrix} b & c \\ e & f \end{vmatrix} = 0, \begin{vmatrix} a & c \\ d & f \end{vmatrix} = 0$

$\frac{a}{d} = \frac{b}{e}, \frac{b}{e} = \frac{c}{f}, \frac{a}{d} = \frac{c}{f}, \boxed{\frac{a}{d} = \frac{b}{e} = \frac{c}{f}} \quad \text{--- (1)}$

$\begin{vmatrix} d & e \\ g & h \end{vmatrix} = 0, \begin{vmatrix} e & f \\ h & i \end{vmatrix} = 0, \begin{vmatrix} d & f \\ g & i \end{vmatrix} = 0, \frac{d}{g} = \frac{e}{h}, \frac{e}{h} = \frac{f}{i}, \frac{d}{g} = \frac{f}{i}, \boxed{\frac{d}{g} = \frac{e}{h} = \frac{f}{i}} \quad \text{--- (2)}$

From (1) (2) (3) $\underline{a:b:c = d:e:f = g:h:i}$

(Case I) $\{a, b, c\} = \{d, e, f\} = \{g, h, i\} \rightarrow {}^6C_3 \times 3! = \frac{6 \times 5 \times 4}{6} \times 6 = 2 \times 6 = 120$

(Case II) $\{a, b, c\} = \{1, 2, 3\}, \{g, h, i\} \neq \{d, e, f\} = \{2, 4, 6\} \rightarrow {}^3C_2 \times 3! \times 2 = 3 \times 6 \times 2 = 36$

4. If $f(x) = \lim_{n \rightarrow \infty} \frac{[(n+x)(n+2x) \dots (n+2nx) \left(\tan^2 \frac{\pi}{2n} \cdot \tan^2 \frac{2\pi}{2n} \dots \tan^2 \frac{(n-1)\pi}{2n} \right)]^{\frac{1}{2n}}}{n} \forall x \neq 0$, is continuous at $x=0$ then find $f(0)$ $\rightarrow$ (1)

$$f(x) = \lim_{n \rightarrow \infty} \left[\frac{(n+x)(n+2x) \dots (n+2nx)}{n^{2n}} \right]^{\frac{1}{2n}} \left[\tan^2 \frac{\pi}{2n} \dots \tan^2 \frac{(n-1)\pi}{2n} \right]^{\frac{1}{2n}}$$

$$\sqrt[n]{f(x)} = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{x}{n}\right) \left(1 + \frac{2x}{n}\right) \dots \left(1 + \frac{2nx}{n}\right) \right]^{\frac{1}{2n}} \left[\tan^2 \frac{\pi}{2n} \dots \tan^2 \frac{(n-1)\pi}{2n} \right]^{\frac{1}{2n}}$$

$$y = f(x) = \lim_{n \rightarrow \infty} \prod_{x=1}^{2n} \left(1 + \frac{x}{n}\right)^{\frac{1}{2n}} \prod_{x=1}^{n-1} \left(\tan^2 \frac{x\pi}{2n}\right)^{\frac{1}{2n}}$$

$$\ln y = \frac{1}{2} \int_0^2 \ln(1+tx) dx + \int_0^1 \ln \tan\left(\frac{t\pi}{2}\right) dt \quad \frac{t\pi}{2} = p \quad dt = \frac{2}{\pi} dp$$

h

$$\frac{2}{\pi} \int_0^{\frac{\pi}{2}} \ln \tan p \, dp \quad (Kims)$$

$$\ln y = \frac{1}{2} \int_0^2 \ln(1+tx) dt$$

$$1+tx = \alpha$$

$$x dt = d\alpha$$

$$\ln y = \frac{1}{2x} \int_1^{1+2x} \ln \alpha d\alpha$$

$$\ln y = \frac{1}{2x} \left[\alpha (\ln \alpha - 1) \right]_1^{1+2x}$$

$$\ln y = \frac{1}{2x} \left[(1+2x) [\ln(1+2x) - 1] - 1 \right]$$

$$\ln y = \frac{(1+2x) \ln(1+2x) - (1+2x) - 1}{2x}$$

$$\lim_{x \rightarrow 0} \frac{(1+2x) \ln(1+2x) - (1+2x) - 1}{2x}$$

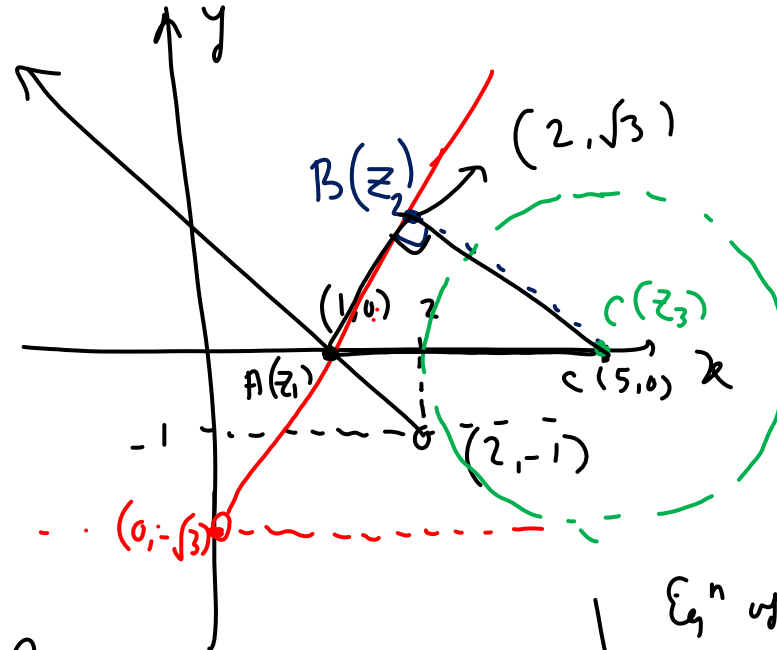
$$\ln y = 0$$

$$y = e^0 = 1$$

5. Let $A(z_1)$ be the point of intersection of curves $\arg(z - 2 + i) = \frac{3\pi}{4}$ and $\arg(z + \sqrt{3}i) = \frac{\pi}{3}$ in the argand plane, $B(z_2)$ be the point on $\arg(z + \sqrt{3}i) = \frac{\pi}{3}$ such that $|z_2 - 5|$ is minimum and $C(z_3)$ be the centre of circle $|z - 5| = 3$, then the area of triangle ABC is

Sol

$(5, 0)$



Eg of line AB

$$y + \sqrt{3} = \tan \frac{\pi}{3} (x - 0)$$

$$y + \sqrt{3} = \sqrt{3}x \quad (1)$$

Eg of BC

$$y - 0 = -\frac{1}{\sqrt{3}} (x - 5)$$

$$\sqrt{3}y = 5 - x \quad (2)$$

$$\text{①} \times \sqrt{3}$$

$$\sqrt{3}y = -3 + 3x \quad (3)$$

$$-3 + 3x = 5 - x$$

$$x = 2$$

Area of ΔABC

$$\hookrightarrow \frac{1}{2} \times 2 \times \sqrt{12}$$

$$\Rightarrow \sqrt{12} \Rightarrow 2\sqrt{3} \Rightarrow 2 \times 1.73$$

$$\Rightarrow \boxed{3.46}$$

6. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar unit vectors equally inclined to one another at an angle of $\frac{\pi}{3}$. If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ then $2(p^2 + q^2 + r^2)$ is equal to (where p, q, r are scalars)

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$$

$$\vec{a} \cdot \vec{b} = \frac{1}{2}, \quad \vec{b} \cdot \vec{c} = \frac{1}{2}, \quad \vec{c} \cdot \vec{a} = \frac{1}{2}$$

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$$

dot with $\vec{c}$

$$[\vec{a} \vec{b} \vec{c}] = \frac{1}{2} + \frac{1}{2} + r \quad \text{--- (1)}$$

dot with $\vec{a}$

$$[\vec{a} \vec{b} \vec{c}] = p + \frac{1}{2} + \frac{1}{2} \quad \text{--- (2)}$$

dot with $\vec{b}$

$$0 = \frac{1}{2} + q + \frac{1}{2} \quad \text{--- (3)}$$

$$(1) = (2)$$

$$\frac{1}{2} + \cancel{q} + r = p + \cancel{q} + \frac{1}{2}$$

$$p = r, \quad p = -q$$

$$p = [\vec{a} \vec{b} \vec{c}]$$

$$2(p^2 + q^2 + r^2)$$

$$\Rightarrow 2 \times 3 \times [\underline{\underline{\vec{a} \vec{b} \vec{c}}}]^2 \Rightarrow \cancel{2} \times 3 \times \cancel{\frac{1}{2}} \Rightarrow \boxed{3}$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = 1 \left[1 - \frac{1}{4} \right] - \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right] + \frac{1}{2} \left[\frac{1}{4} - \frac{1}{2} \right]$$

$$\Rightarrow 1 - \frac{1}{4} - \frac{1}{4} + \frac{1}{8} + \frac{1}{8} - \frac{1}{4}$$

$$\Rightarrow 1 - \frac{1}{2} \Rightarrow \boxed{\frac{1}{2}}$$

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